



Republic of Zambia

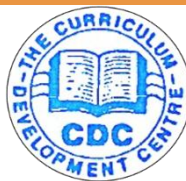
Ministry of Education

MATHEMATICS I

TEACHING MODULE

FORM 2

TERM 1 & 2



Developed by the Curriculum Development Centre

Lusaka

2026

© Curriculum Development Centre, 2026.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without prior written permission of the copyright owner.

Vision

Quality, lifelong education for all which is accessible, inclusive and relevant to individual, national and global needs and value systems.

PREFACE

The Mathematics I Form2 Teaching Module has been developed to provide guidance to the teacher on how to deliver competence-based lessons which are age-appropriate for learners at this level. The teaching module is aimed at providing quality education that is aligned with the Competence-Based Curriculum and 21st Century Skills. It also aims to facilitate the building of knowledge, skills, values and positive attitudes that are aimed at enabling children to live and grow into productive and useful members of their communities and the Zambian society.

The guide provided in this teaching module is aimed at facilitating the acquisition of knowledge, competences and values to nurture a learner who is creative, critical and an analytical thinker. The suggested learning activities are designed to offer learners hands-on and minds-on experiences through manipulative tasks. Thus, the learners will develop knowledge, desirable lifelong skills, values and positive attitudes needed for their personal and national development.

It is hoped that the guidance provided in this module will make learning more meaningful and enjoyable as it is highly activity oriented and allows for a smooth transition into the Constructivist way of learning.

Kelvin Mambwe (Dr)
Permanent Secretary- Educational Services
MINISTRY OF EDUCATION

ACKNOWLEDGEMENT

Many thanks go to individuals, institutions and organizations that participated in the successful development of this Module. These include teachers, lecturers from colleges and public universities in Zambia, Senior education standards officers and experts from Directorates in the Ministry of Education.

I also recognise the commitment and hard work of Staff at the Curriculum Development Centre in ensuring that this module is successfully developed.

Charles Ndakala(Dr.)
Director- Curriculum Development
MINISTRY OF EDUCATION

AUTHORS

Shelly MizingaSikwale

Curriculum Specialist -Mathematics, Directorate of Curriculum Development.

Naomy Kalabwa

Curriculum Specialist Mathematics, Directorate of Curriculum Development.

Patience M.Chisanga

Senior Education Standards Officer, Mathematics (Copperbelt Province)

Diaphord L. Shamasamu

David Livingstone Secondary School, H.O.D (Mathematics)

Tafeni Spiwe

Mkushi Day Secondary School, Subject Teacher (Mathematics)

Table of Contents

PREFACE	iv
ACKNOWLEDGEMENT	v
INTRODUCTION	viii
PURPOSE OF THE MODULE	viii
HOW TO USE THE MODULE	viii
TOPIC 1: COORDINATE GEOMETRY.....	10
Sub-Topic 1: Cartesian Plane	11
Sub-Topic 2: Straight lines on the cartesian plane.....	28
TOPIC 2: RELATIONS AND FUNCTIONS.....	69
Sub-Topic 1: Relations.....	70
Sub-Topic 2: Functions.....	79
TOPIC 3 : ALGEBRA	110
Sub-Topic: Basic Processes of Algebra:.....	111
TOPIC 4: POLYGONS.....	149
Sub-Topic 1: Regular and Irregular Polygons	151
References	173

INTRODUCTION

The Mathematics I Form 2 Teaching Module has been developed to guide teachers on how to facilitate Competence-Based lessons at this level. The module aims to provide quality education that is aligned with the Competence-Based curriculum and 21st Century Skills, that is developing knowledge, skills, values and positive attitudes that are aimed at enabling learners to live and grow into productive and useful members of their communities and the Zambian society.

PURPOSE OF THE MODULE

The purpose of the module is to illustrate teaching and learning activities that:

- a) are consistent with teaching and learning in a competence-based curriculum;
- b) show processes that allow achievement of 21st Century Skills such as creativity, collaboration, communication and critical thinking;
- c) Illustrate what activity-based and learner-centred teaching and learning could look like.

HOW TO USE THE MODULE

This module provides **suggested learning activities and teaching strategies** that enhance learners' mathematical competences. The suggested activities are designed to offer learners hands-on experiences through manipulation of tasks that are interactive and learning through practice. We should bear in mind that the desired competences and 21st century skills will NOT be achieved by the topics 'covered' but by the manner in which teaching and learning activities are carried out. For example, asking learners to give reasons for their answers, whether or not the answer is correct; to plan for solving a problem; to consider alternative

ways of solving a problem even if the first method yields a correct answer and to consider which of the methods is better; to acknowledge that even in mathematics, more than one correct answer is possible.

Activities in the Module are meant to be incorporated in the teacher's lesson plan. The activities are NOT meant to replace the teacher's lesson plan. The topics and activities in the Module are as specified in the 2024 Mathematics Form 1 to 4 Syllabus. There might be terms used in the Module that are not exactly the same as those in the syllabus, but this is in the interest of using language that might be more learner friendly. Teachers are encouraged to vary the environment when conducting lessons in order to create a good atmosphere for learning and teaching which is learner friendly.

Each topic in the Module begins with a *key question*. A key question seeks to stimulate learners' curiosity and motivate them to learn. Teachers may create their own key questions that provoke their learners' curiosity and motivation.

ICONS USED IN THE MODULE

This module utilises icons as graphics to represent instructions, enhancing the learning experience for learners, icons categorize and organise instructions, making navigation easier for teachers.



Key terms



Content Tips



Assessment



Discussion



Activity

TOPIC 1: COORDINATE GEOMETRY

INTRODUCTION

Coordinate Geometry is a branch of Mathematics that helps us locate and describe the position of points on a plane using ordered pairs (x,y) . It connects Algebra and Geometry, allowing us to draw graphs, measure distances, and understand shapes more clearly. Beyond the classroom, Coordinate Geometry plays a vital role in everyday life. It is used in maps and Global Positioning System (GPS) to find locations, in architecture and engineering to design buildings and bridges, in computer graphics to create animations and video games, and even in navigation systems for cars and airplanes.

Learning Coordinate Geometry equips learners with practical tools to understand both Mathematics and the world around them. It fosters problem-solving skills, spatial awareness, and the ability to connect abstract concepts with real world applications-making it an essential foundation for further studies in Mathematics, Science, and Technology.



Key terms

Geometry, cartesian plane, Origin, x-axis, y-axis, Coordinate, Ordered pairs, point, Mid- point, Gradient.

General competence(s):

Analytical thinking, Critical thinking, Communication, Problem solving, Collaboration. Digital Literacy.



Key question



*How can you describe
your Location?*

Sub-Topic 1: Cartesian Plane

Introduction

The Cartesian plane is a two-dimensional (2D), infinite plane formed by a horizontal x-axis and a vertical y-axis. These axes intersect at a central point called the origin. The Cartesian plane helps us visualize and study relationships between two quantities. By plotting points, drawing lines, or curves, we can understand patterns and connections in both mathematics and real-life situations.

Specific competence: Use the concept Cartesian plane to solve problems in real life situations

Learning activity 1: Exploring ways of describing a location (grids, maps, sitting arrangements...)



Activity 1.1: Location in the classroom

Instructions:

- Ensure that learners are in clearly defined rows and columns.
- Ask learners to describe their own location using rows and columns.
- Ask learners to find a friend's location using rows and columns.



Content Tips

- Rows and columns can be numbered to make the location description easier.
- For instance, in activity 1.1 learners can say Jane is seated on row 3 column 6



Activity 1.2: Describing and Finding Locations in Our Environment

Instruction:

In small groups learners should perform the following assignment:

Scenario

Your school is receiving new learners. They need help finding places like the classroom, staff room, toilets, and playground. Your group must create a **system to describe locations clearly**.

Task 1: Create a visual representation of a section of the school (e.g. a grid map of the classroom or school **or** a simple map with landmarks ...)

Task 2: Describe Locations in 3 Different Ways

Choose at least 3 places within the school compound (e.g., door, desk, tree, office) and describe their location using any or all the following:

- i. Relative position (e.g., "Next to..., behind..., in front of...")
- ii. Directions / landmarks (e.g., "Walk straight, turn left at the office...")
- iii. Grid (if using a grid) (e.g., "(C,4)")

Task 3: Create a Challenge for Another Group

- i. Write 3 clues/instructions describing a hidden location

- ii. Another group must follow your instructions and find it



Task 4: Reflection

Answer these questions as a group:

- i. Which method of describing location was the most accurate? Why?
- ii. Which method was easiest to understand? Why?
- iii. When would you use each method in real life?



Content tips:

- Learners will explore different ways of describing location
- The following rubric can be used to assess learners

Criteria	4 (Excellent)	3 (Good)	2 (Basic)	1 (Needs Support)
Representation (Map/Grid)	Clear, accurate, well-organized	Mostly clear, minor errors	Some errors, partially clear	Incomplete or unclear
Clarity of Instructions	Very clear and easy to follow	Mostly clear	Some confusion	Hard to follow
Application (Challenge Task)	Other group easily finds location	Found with minor difficulty	Found with help	Not found

Criteria	4 (Excellent)	3 (Good)	2 (Basic)	1 (Needs Support)
Collaboration	Excellent teamwork and participation	Good teamwork	Uneven participation	Poor teamwork

Learning activity 2: Exploring the Cartesian Plane (XOY plane) and label it (x-axis, y-axis and origin)



Activity 2.1: Cartesian plane construction

Instructions:

- Learners draw a horizontal number line.
- Learners draw a vertical number line.
- Learners draw both the horizontal and vertical number line by intersecting them at an appropriate point. Learners justify their choice of point of intersection.



Content tips:

- A cartesian plane is a system used to locate points on a plane.
- In activity 2.1 learners explore how to create this system.
- Learners may intersect the two number lines at any point of their choice. However, guidance should be given that the correct point of intersection of the two lines is the point where both axes are zero because this is the starting point of the two number lines.
- The origin is where the x- axis and y- axis are both zero.
- The x axis is the horizontal number line.
- The y axis is the vertical number line.
- The two lines divide the cartesian plane into four parts called quadrants.
- It acts as a reference point for locating all other points on the cartesian plane.

Learning activity 3: Plotting and reading ordered pairs on the XOY-plane, (x,y) starting with the x-coordinate followed by the y-coordinate



Activity 3.1: Human Coordinate Plane

Instructions (Whole Class)

- Mark out a large grid on the floor or ground
- Learners to draw and label the Origin, x- axis and y- axis

- One learner (learner A) to stand on the origin while the teacher shouts out step movements (e.g. 2 steps to the right and 3 steps up) and learner A moves to that point
- Place learners in groups
- Each group draws a grid and chooses two learners
- Write two instructions on the board:
 - **Learner A:** Move 3 steps right, then 2 steps up
 - **Learner B:** Move 3 steps up, then 2 steps right
- Each group follows the instructions and places the two learners (A and B) on the correct spot.



Discussion: each group to discuss the following and present to the class for consolidation

- What did you notice?
 - Why are the final positions different?
 - What did you do differently in A and B?
 - Which movement did you do first each time?
- After presentation learners discuss the following (after the order pair rule is discussed)
 - Points on the cartesian plane are represented using ordered pairs in which we start with the x value followed by y value, and is written as (x,y) .
 - i. What would happen if we swapped x and y values in describing a point?
 - ii. Why is it important for the order to remain the same on maps and graphs?



Content tips:

- **Ordered Pairs** are a way of representing an exact position of a point on a grid.
- Each location on a plane can be described using an ordered pair, starting with x value followed by y value, and is written as (x, y) .
- Switching the order gives a different location. (Eg
 - o $(2, 5) \rightarrow$ 2 across, 5 up
 - o $(5, 2) \rightarrow$ 5 across, 2 upthese will give two different points



Activity 3.2: Project Title: "Our School on the Grid"

Project Goal: Learners will design a scaled grid map of the school compound, assign coordinates to key places and use directions to describe and locate positions.

Instructions

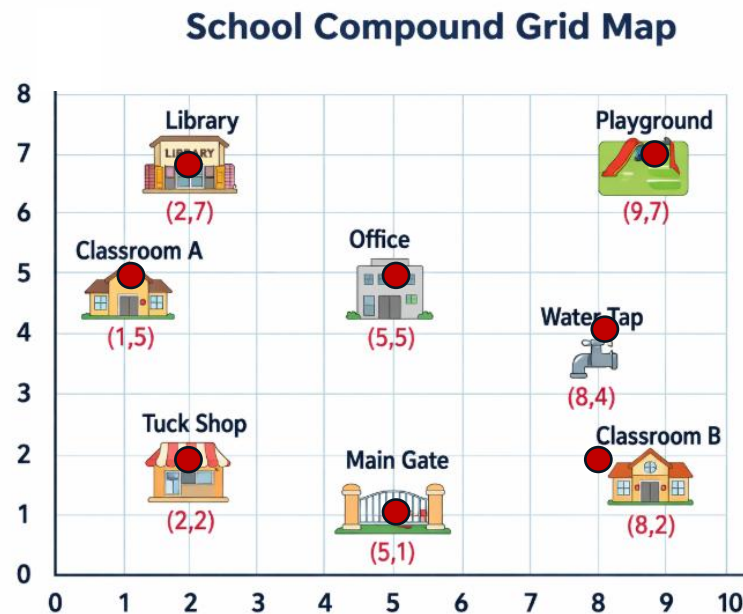
- Place learners in small groups
- Learners follow these steps

Procedure:

i. Collect Data

- Walk around the school compound with your group.

- Identify important places such as the library, office, playground, classrooms, water taps, and other structures.
- Estimate the distance between them and make a rough sketch of where they are located.



ii. Design the Map:

- Draw a grid map of the school compound using a suitable scale.
- Assign coordinates to each structure you identified.

- iii. Describe Locations:
Write short descriptions of the location of each structure using coordinates.
Example: *"The library is at (5,3)."*
- iv. Create Directions
Write directions that someone can follow to move from one place to another on your map.
Example: *"Move 3 units east and 2 units north."*
- v. Present Your Work
Each group will present their map to the class for peer review and consolidation.

Optional Extension: Add a "treasure hunt" challenge where learners hide clues and use coordinates to find them.



Activity 3.3: Mall/Market Directory

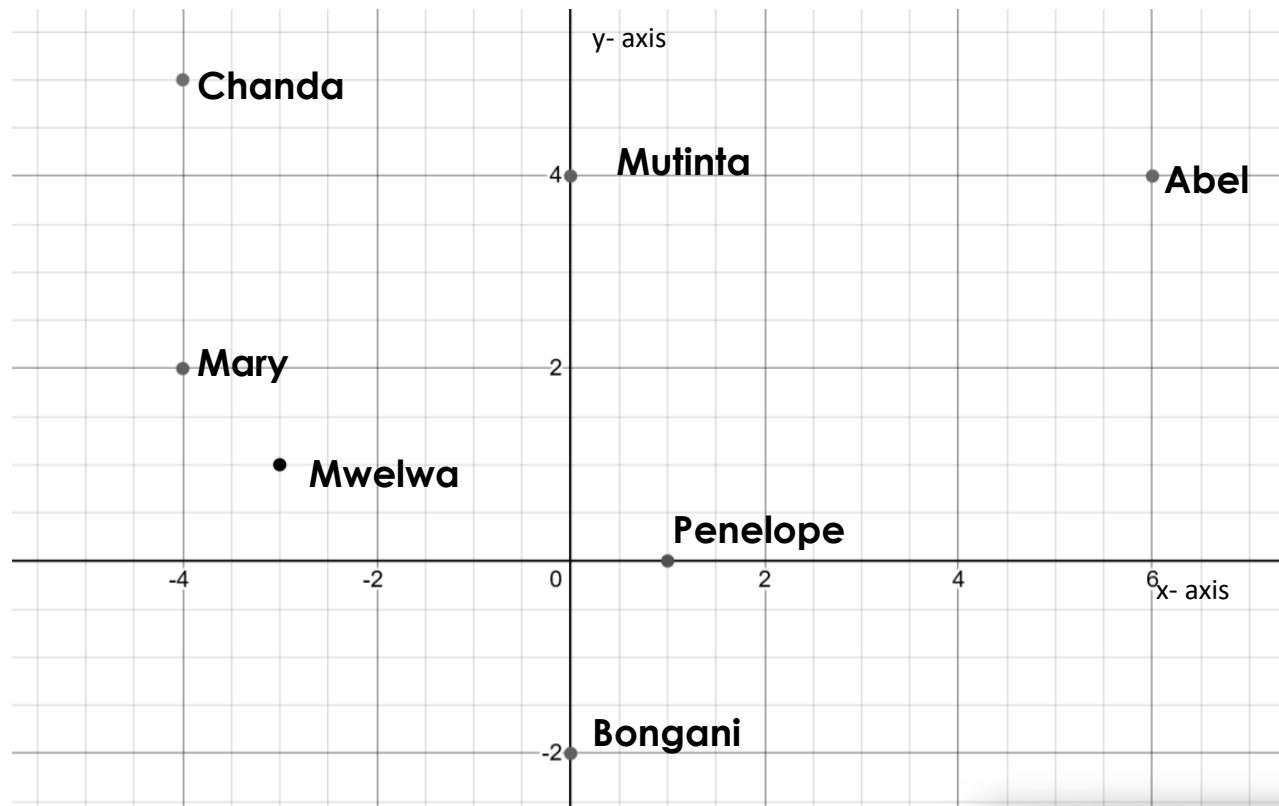
- Individually, learners to design a mall/ market floor plan on a grid. (only 1st quadrant), where Shops/market stands have coordinates.
- learners:
 - Plot shop/market stands locations
 - Read coordinates of given shops/market stands
 - Explain what each of the coordinates represent



Activity 3.4: Class grid

- Learners to answer the following questions individually

A class grid has been drawn on the floor of Chanda's class. This has been illustrated on the following cartesian plane. From the illustration, using ordered pairs, state the positions of learners shown.



Learning activity 4: Joining points on the XOY plane to come up with different shapes (triangles, quadrilaterals ...)



Activity 4.1: Shape formation

- Begin the activity in small groups and after consolidation give the same activity to individual learners
- Put learners in small groups
- Give each group a set of coordinates (can be on paper or cards)
- Ask learners to plot the points given and join them to form shapes.
- Learners identify the shapes formed and give properties that aided in the identification.



Content tips:

- Learners to give properties that assisted in the identification of the shape (e.g. the number of sides, size of angles)



Activity 4.2 : Town block Design (Urban Planning)

Instructions :

- Learners perform the following task individually.

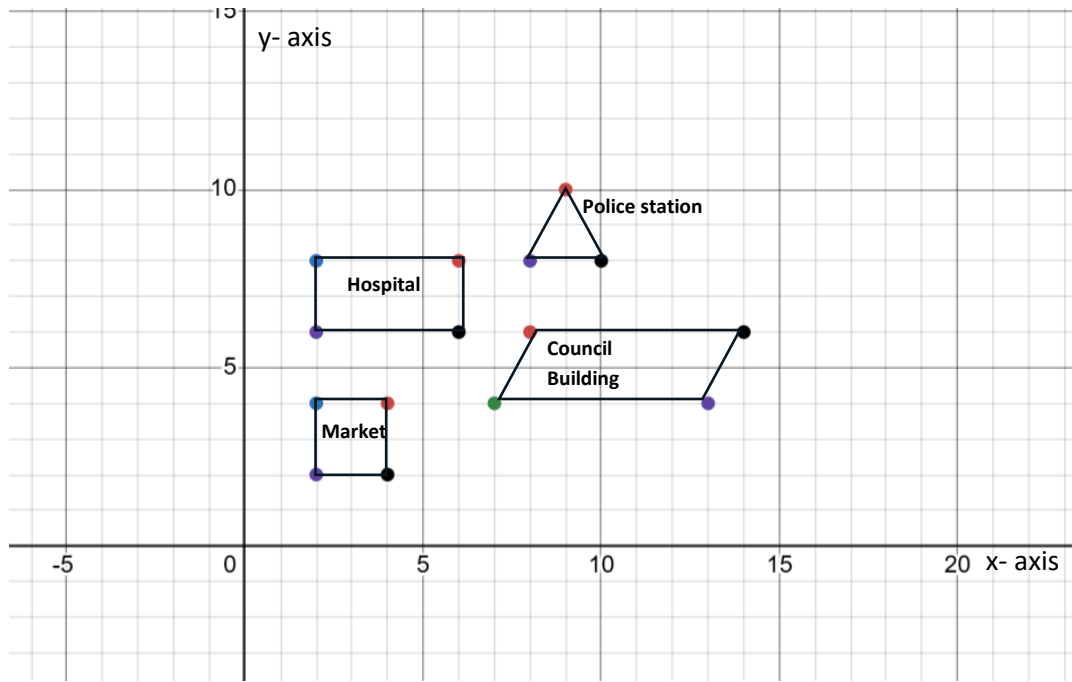
Scenario: You are a town planner, and your task is to design a simple town using a Cartesian plane.

- i. Draw a Cartesian plane and use only the first quadrant (where both x and y are positive).
- ii. Place the following buildings on the grid using the given shapes:
 - o Hospital – Rectangle
 - o Police Station – Triangle
 - o Council Building – Parallelogram
 - o Market – Square
- iii. For each building:
 - o Draw the shape on the grid.
 - o Make sure all corners lie on grid intersection points.
- iv. Write down the coordinates of each corner (vertex) of every building.
- v. Label each building clearly on your diagram.
- vi. Make sure your buildings do not overlap and are neatly arranged.
- vii. Explain why you chose the positions of your buildings (e.g., why the market is near the hospital or police station).



Content tips:

- All the 4 buildings should be well spaced, with corners (vertices) accurately placed on a line intersection. For instance, the buildings may be placed as shown in the following:



Learning activity 5: Creating different patterns using the cartesian plane



Activity 5.1: Designing repeated patterns (tessellation) using coordinates

Instructions:

- The teacher divides learners into small groups.

- Each group receives graph paper, rulers, pencils, and coloured pencils.
- Each group analyses the scenario and answers the questions that follow

Scenario Title:*The School Hall Floor Design Project*

Your school plans to renovate the floor of the assembly hall. The school wants a unique floor tile design that uses repeating patterns similar to those seen in floor tiles, fabric prints, or traditional textiles. The school has invited the mathematics club to help design these patterns using graph paper and coordinate points.

Task:

- Plot Points: On your graph paper, plot several points using coordinates in the form (x, y) .
- Create a Shape: Join the points using straight lines to form a shape. You can create shapes such as a triangle, square, hexagon, or any other shape you like.
- Repeat the Shape : Copy your shape and move it:
 - Horizontally (left or right), or
 - Vertically (up or down)
 - Do not change the size or shape, just slide it across the grid to create a repeating pattern (tessellation).
- Form a Pattern: Keep repeating the shape until your grid has a clear pattern covering a large area.
- Add colour: colour your shapes to make the pattern neat, clear, and attractive.
- How are repeated patterns (tessellations) used in real life?

Note:

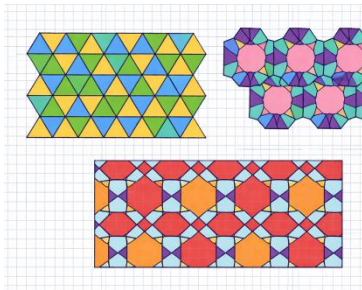
- Keep your shape the same size when moving them.
- Make sure shapes fit together without gaps or overlaps

- **Extension Option :** Using software like GeoGebra or Desmos, Learners can experiment with plotting lines to create shapes and generate artistic patterns.



Content tips :

- Example of tessellations



- Repeated patterns (tessellations) are used to:
 - Cover large areas neatly and attractively
 - Add style and identity to textiles
- The following rubric can be used to grade the task

Criteria	Excellent (4)	Good (3)	Basic (2)	Needs Improvement (1)
Plotting of Points	Points are accurately plotted on correct	Most points are correct with few	Some points are correct but	Points are mostly incorrect or

Criteria	Excellent (4)	Good (3)	Basic (2)	Needs Improvement (1)
	coordinates.	errors.	several errors.	missing.
Shape Construction	Shape is correct, clear, and well-formed (e.g., triangle, square, etc.).	Shape is mostly correct with minor mistakes.	Shape is unclear or has errors.	Shape is incorrect or not completed.
Use of Translation	Shape is correctly repeated by sliding (no change in size or orientation).	Shape is repeated with small errors.	Some attempt to repeat but not consistent.	No clear repetition or incorrect movement.
Tessellation Pattern	Pattern covers space fully with no gaps or overlaps.	Pattern mostly complete with few gaps/overlaps.	Pattern incomplete or irregular.	No clear pattern formed.
Neatness and Accuracy	Work is very neat, lines are straight, and grid is used correctly.	Mostly neat with small issues.	Some neatness but inconsistent.	Work is untidy and hard to follow.
Use of Colour	Colour is used effectively to highlight the pattern clearly.	Colour is used but not consistent.	Limited use of colour.	No colour or does not add clarity.

Scoring Guide

- **20–24** → Excellent
- **14–19** → Good
- **8–13** → Basic
- **1–7** → Needs Improvement

Expected standard: The concept of the cartesian plane applied in real life correctly.

Sub-Topic 2: Straight lines on the cartesian plane.

Introduction

A straight line on the cartesian plane is formed when several points are connected in straight path. Each point on the line has coordinates written in the form (x, y) . straight lines are important because they help us understand how two variables are related and how one quantity changes with another. In real life straight lines on a cartesian plane can be used to represent situations such as distance against time or Cost against number of items.

Specific competence: Use straight lines to solve problems in real life

Learning activity 1: explore the cartesian plane and its usefulness in real life (cost benefit analysis, demand and supply...)



Activity 1.1 : Cost Benefit Analysis on a Cartesian Plane

Instructions

- In pairs learners perform the task

Scenario

Your community wants to start a development project but must choose the one with the best benefit for the cost. Using a Cartesian plane, plot the costs and benefits of each project, compare them, and decide which project the community should choose.

Step 1:

Draw a Cartesian plane and label:

- x-axis → Cost (in thousands of Kwacha)
- y-axis → Benefit (in thousands of Kwacha)
- Origin (0,0)

Step 2: Plot Project Data

The teacher provides the following information:

Project	Cost (K'000)	Benefit (K'000)
Borehole	3	8
School Garden	2	6
Poultry Project	5	9
Fish Pond	4	7

Step 3: Draw Comparison Lines: draw straight lines from the origin (0,0) to each project point.

Step 4: Answer the following questions

- From the graph which project:
 - i. has the highest benefit?
 - ii. has the lowest cost?
 - iii. gives the best benefit compared to cost?
 - iv. would you recommend for the community? Why?
 - v. What do the straight lines from the origin help you see that the table does not show clearly?
 - vi. What is the advantage of using a graph when making real-life decisions like this?



Content tips:

- Some of the advantages of using a cartesian plane are :
 - It makes comparison faster and clearer
 - You can see relationships visually
 - It helps identify the best option quickly
 - It shows value (benefit per cost) through steepness
 - Steeper lines = better benefit compared to cost
 - Flatter lines = lower benefit compared to cost

Learning activity 2: Exploring the straight line on the cartesian plane



Activity 2.1: Connecting the Dots

Materials Needed

- Graph paper, Rulers, Pencils, Coloured pencils

Task:

1. Draw a Cartesian plane on graph paper.
2. Plot the following points:

Point	Coordinates
A	(1,2)
B	(5,6)
C	(2,5)
D	(6,5)
E	(1,3)

Point	Coordinates
F	(1,7)

3. Connect points A and B with a ruler to form a straight line.
4. Connect points C and D with a ruler to form another straight line.
5. Connect points E and F with a ruler to form another straight line.
6. State the coordinates of two other points on each of the lines.
7. If you extend line AB, which other point will it pass through?
8. Can you predict a point that will NOT lie on the line? Explain why
9. What can you observe about line CD and EF?



Content Tips:

- A straight line passes through all points in a single direction.
- The line can be horizontal, vertical, or slanted depending on the points.
- Online Graphing Tools such as Desmos can be used to give visual demonstrations

Steps in Desmos

- i. Open the website <https://www.desmos.com/calculator>

- ii. Type the line equation directly in the input box:

$$y = 2x + 3$$

(at this point do not give the learners the equation, it is just for demonstration purposes)

- iii. The line will appear immediately on the graph.
iv. You can zoom in/out, move axes, and add points to check values.
v. Optional: Add labels for points, intercepts, and gradient



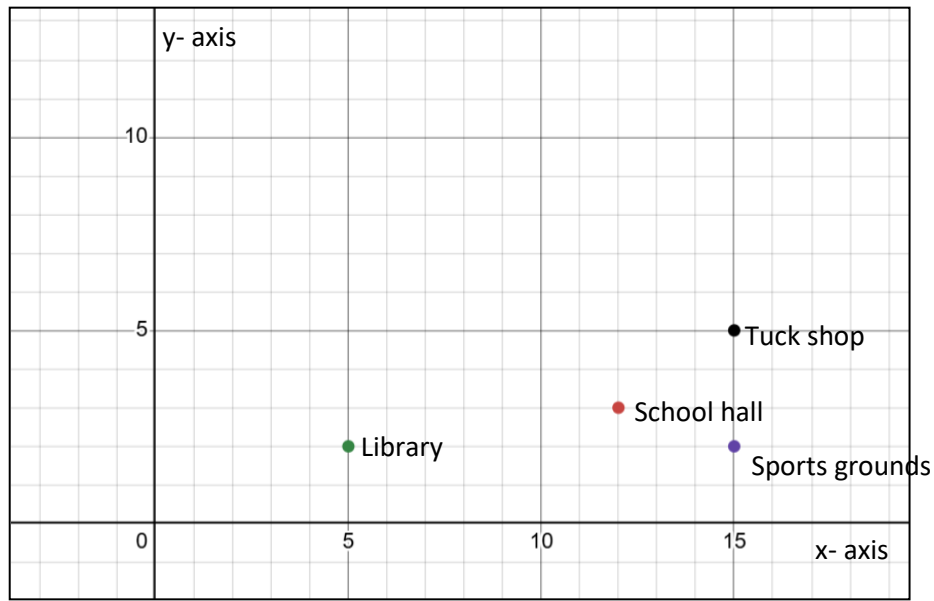
Activity 2.2: Length of a straight line

Instructions:

- Learners analyse the scenario in pairs and answer the questions that follow

Scenario:

The school wants to build a new footpath from the library to the tuck shop so that learners can move faster during break time. Mary's class has been tasked to determine the shortest distance between the two points. After seeing the grid map James suggests that the path should be from the tuck shop to sports ground to the library. Robert on the other hand, wants the path to go from tuckshop through the school hall to the library and Mary wants it to be a direct path from the tuckshop.

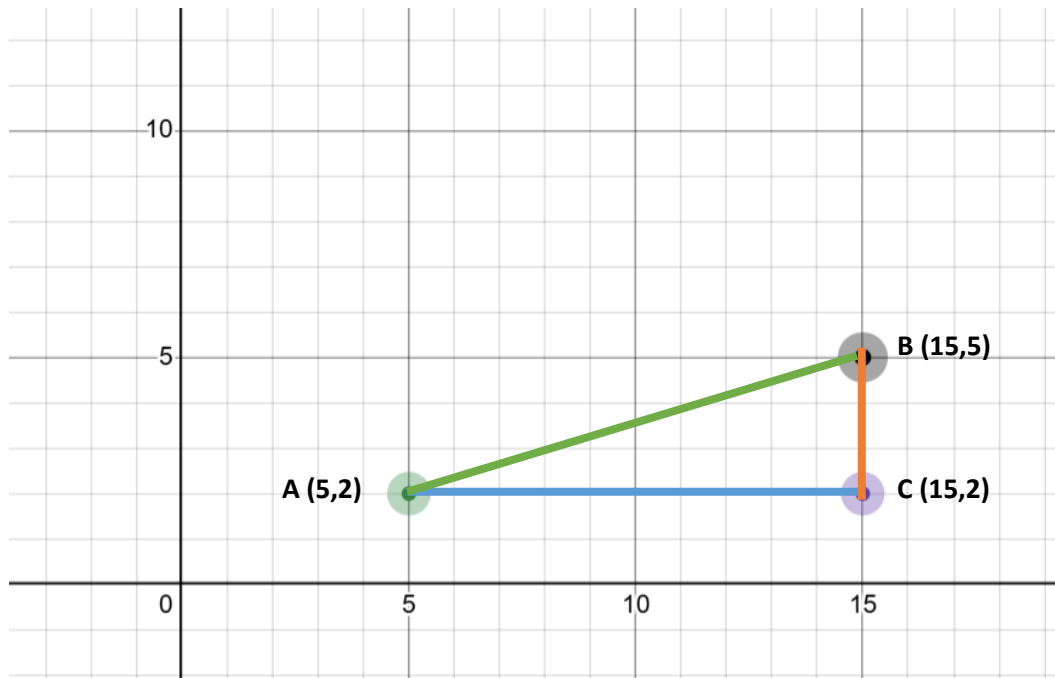


- i. Determine which of the following suggestions is the correct one and give justification for your answer
- ii. How can we use the cartesian plane to find the length of a line? Can Pythagoras theorem be used and how?



Content tips:

- The length of a line can be determined by
 - i. Horizontal lines
If two points have the same y-coordinate (x_1, y) and (x_2, y) , the line is horizontal, $Distance = (x_1 - x_2)$
 - ii. Vertical Lines
If two points have the same x -coordinate (x, y_1) and (x, y_2) , the line is horizontal, $Distance = (y_1 - y_2)$
 - iii. Slanted line
- A line is slanted when both x and y are different. Pythagoras theorem is used to derive a formula for the line



- The generalised form of Pythagoras theorem to find the length of a line AB where A(x_1, y_1) and B(x_2, y_2) is

$$\text{length } AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



Activity 2.3: Distance between two points

Instructions:

- Learners should answer the following questions individually.
1. The school wants to build a new path between the library (2,2) and the tuck shop (8,5).
 - i. Using the coordinates, find the shortest distance between the library and the tuck shop.
 - ii. If a learner walks about 1 meter per step, approximately how many steps will they take to walk from the library to the tuck shop?
 2. A local borehole is at (3,1) and the market is at (7,6) on a village map grid. How far is it from the borehole to the market?
 3. A football field has a goalpost at (4,2) and another at (10,8).
 - i. Find the distance between the two goalposts.

- ii. Which path is shorter: walking along the field edges or walking straight across?
4. A new road is to be built from the police station (1,1) to the hospital (6,5) in town. Calculate the length of the road using the coordinates.
5. A weather station records the temperature over a day in winter. The x-axis represents hours of the day, and the y-axis represents temperature in °C.

Hour (x)	Temperature (y)
6	-2
10	3
14	5
18	-5

- i. Plot the points on a Cartesian plane:
 - A(6, -2), B(10, 3), C(14, 5), D(18, -5)
- ii. Draw the line segments connecting the points in order of the day. (that is A to B , B to C)

iii. Calculate the length of the line between:

a) A and B

b) C and D

iv. What can the graph tell us about how temperature changes between the different times.

Learning activity 3: Determining the mid-point of two points.



Activity3.1: Middle point discovery

Instructions:

- In small groups learners should analyze the following scenario and answer the questions that follow

Scenario: The constituency development committee wants to sink a bore hole at a point that is exactly halfway between the school and the market so that it is easy for everyone to reach.

Use the cartesian plane to find the point that is halfway between the two locations and state its coordinates, given that the school is at $(2,4)$ and the market is at $(8,6)$.

Note:

If the learners are not able to generalize the a way of determining the midpoint the teacher can ask the following guiding questions to prompt the derivation.

- i. Compare the x coordinate of the two given points with the x coordinate of the half way point.
- ii. Compare the y coordinate of the two given points with the y coordinate of the half way point.
- iii. Using your observations suggest the formula for calculating the half way point

**Content tips:**

- Using any method (counting squares, measuring or calculating), learners may be able to determine the point that lies exactly halfway between the school and the market.
- The halfway point between two points is called the mid-point
- The mid-point can be determined using the length of the line or using the midpoint formula. Given two points A (x_1, y_1) and B (x_2, y_2) the midpoint is given by

$$x_{\text{midpoint}} = \frac{x_1 + x_2}{2}$$

and

$$y_{\text{midpoint}} = \frac{y_1 + y_2}{2}$$

$$\text{midpoint coordinates} \rightarrow \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$



Activity 3.2: Midpoint

Instructuins:

- Learners answer the following questions individually;
- 1. John lives at $(2,3)$ and Mary lives at $(8,7)$ on a town grid map. They decide to meet at a place that is exactly halfway between their homes. Find the coordinates of their meeting point.
- 2. Two villages are located at $(1,5)$ and $(9,5)$ on a map. A bus company wants to place a bus stop exactly halfway between the villages. Find the coordinates of the bus stop.
- 3. Two towns are located at $(0,2)$ and $(10,8)$ on a grid map. A rest station will be built halfway between the towns for travelers. Find the coordinates of the rest station.
- 4. On a town grid map, a school is located at $A(2,6)$ and a clinic is located at $B(x,10)$. The town council builds a water point exactly halfway between the school and the clinic so that both places can access it easily. The water point is located at $(6,8)$. Find the value of x , which represents the x-coordinate of the clinic.

Learning activity 4: Determining the gradient of a straight line on a cartesian plane.



Activity 4.1: Exploring Steepness of a Ramp

Instructions:

- Learners perform the following task in groups.

Materials needed: A wooden board or ruler, Books or blocks, Measuring tape or ruler, Paper and pencils

Procedure:

- Group Formation:** Divide learners into small groups.
- Construct a Ramp:** Each group places one end of a board or ruler on the floor and raises the other end using books or blocks.
- Take Measurements:** Learner's measure:-
 - The vertical height from base, edge of the book/block that is at the bottom of the stack to the point where the book touches the board.
 - The horizontal distance from the base, edge of the book/block that is at the bottom of the stack to the fixed point where the board meets the base (table floor...)
- Record Results;** Learners record their measurements in a table and sketch the results to a given ratio.

- v. **Change the Height;** Add or remove books, measure again and record the new results. Repeat this procedure as many times as possible.
- vi. **Change the horizontal distance;** Move the books closer to or further from the fixed point to reduce or increase the horizontal distance. Repeat this procedure as many times as possible.



vii. **Discussion**

Learners discuss the following questions:

- Which ramp is steeper?
- What happens when the height increases but the distance remains the same?
- What happens when the distance increases but the height remains the same?
- When all the books are removed, what kind of surface is formed? How would you describe its slope compared to the other ramps?

viii. **Group Presentation**

Each group shares their observations with the class for consolidation.



Activity 4.2 : Exploring Steepness of a Ramp on cartesian plane

Instructions:

- **In small groups (3-4) learners perform the following tasks**
- i. On graph paper learners draw a cartesian plane with clearly labeled Origin, x and y axis. Let the
 - horizontal distance be represented on the x-axis.
 - vertical height be represented on the y-axis.
- ii. Learners then draw the line segment representing the board by
 - Plotting the point (horizontal distance, vertical height) on the graph
 - Join the point (0,0) to the point representing the ramp height.
 - The line drawn represents the ramp on the graph
 - Repeat this for all the sketches made (on the same graph)
- iii. Learners then answer the following question:
'From the measurements taken (horizontal distance and vertical height), how can we determine the steepness of each line?'



Content tips:

- the steepness can be determined by a comparison of the **vertical height** in relation to the **horizontal distance**.
- the steepness of a line can be referred to as the gradient of the line.
- More books → larger gradient
- Flatter ramp → smaller gradient
- When all the books are removed and the ramp is completely flat the gradient is zero.



- A vertical line has undefined gradient.

- Gradient is conventionally denoted by the letter m .
- Given any two points on the Straight line A (x_1, y_1) , and B (x_2, y_2) , the gradient (m) of the line can be given by :

$$\text{Gradient } (m) = \frac{y_1 - y_2}{x_1 - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$$

Alternatively

$$m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$$



Activity 4.3 : Sliding Down the ramp

Instructions: in small groups learners perform the following task

Materials: A board or ruler (to act as the ramp), Books to support the ramp, A small object (toy car, marble, or eraser), Graph paper

Step 1: Set Up the Ramp

- Place books on the right side of the table.

- ii. Rest the board on the books so that the left side of the ramp touches the table and the right side is raised.
- iii. Ask learners if the ramp going up or down as we move from left to right?

Step 2: Reverse the Ramp

- i. Move the books to the left side.
- ii. Place the board so the left side is raised and the right side touches the table.
- iii. Ask learners if the ramp going up or down as we move from left to right?

Step 3: Connect to the Graph

- ii. Draw axes on graph paper.
- iii. Mark the starting point (0, height).
- iv. Mark the end point where the ramp touches the table (distance, 0).
- v. Join the two points.
- vi. Determine the gradient.
- vii. Compare the gradient in the first illustration with this one

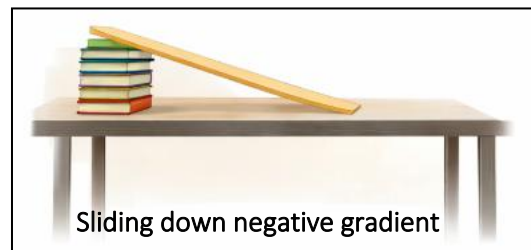
Extension:

- Desmos or Geogebra can be used to illustrate how changing gradient can affect the direction of the line.
- Click or copy the link to view an illustration of changing gradient in desmos for the line $y = mx$. <https://www.desmos.com/calculator/flk3vitug6>
- Press on the play button or slide m line left to right to view the changing gradient.



Content tips:

- When a line goes down as we move from left to the right, the gradient is negative.



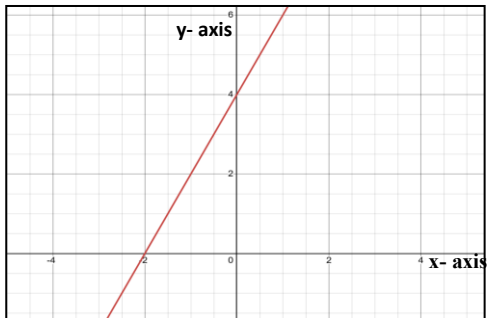


Activity 4.4: Calculating gradient from graph

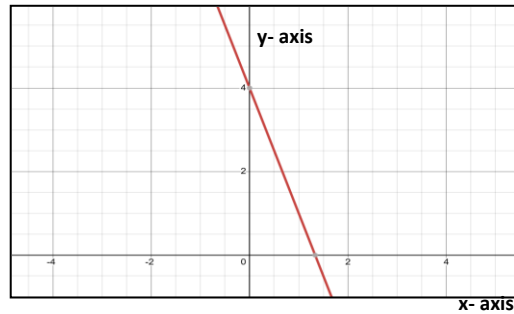
- Learners answer questions given individually:

Calculate the gradient of each of the straight lines given;

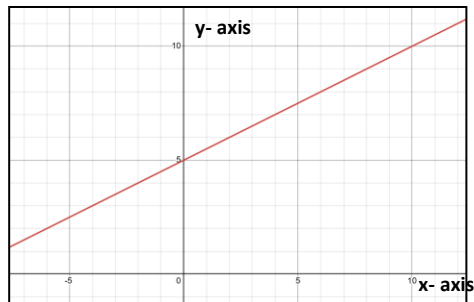
a)



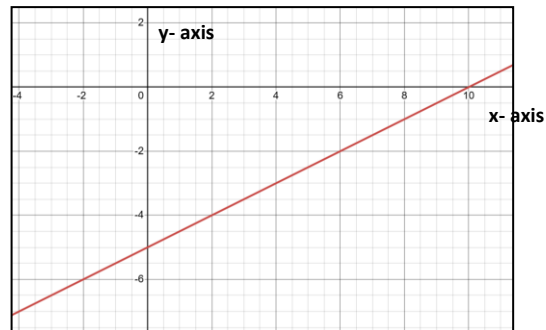
b)



c)



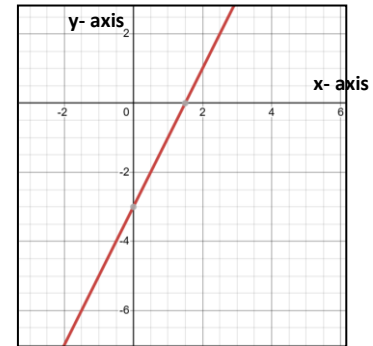
d)



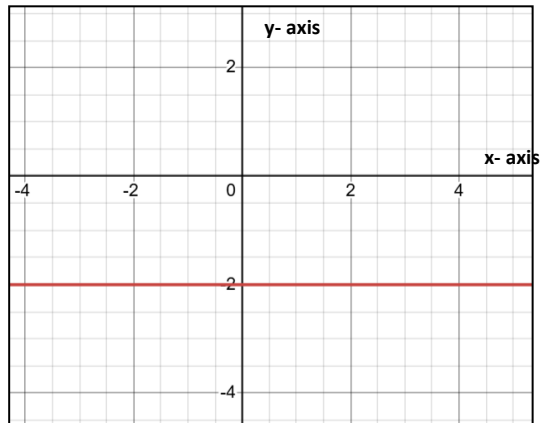
e)



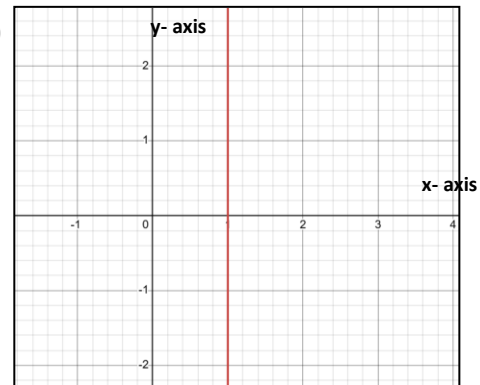
f)



g)



h)





Content tips:

- learners to pick any two points from each line and find the gradient using the gradient formula.
- For instance on graph a) two possible points are (0,4) and (2,0). The gradient will be determined by

$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

$$m = \frac{4 - 0}{0 - 2}$$

$$m = \frac{4}{-2}$$

$$m = -2$$

- Graph g) will give a gradient of zero
- Graph h) gradient is undefined



Activity 4.5: Determining the gradient

- **Instructions:**

Learners answer the following questions individually.

1. A wheelchair ramp is being built at the entrance of your classroom. The ramp rises 1 metre from the ground to the door. The horizontal distance from the start of the ramp to the door is 5 metres.
 - i. What is the gradient (slope) of the ramp?
 - ii. What does this gradient tell you about how steep the ramp is?
 - iii. If the horizontal distance is increased to 10 metres while the height remains 1 metre, how will the gradient change?
 - iv. Which ramp will be steeper, the first or the second? Give reasons for your answer.
2. During the construction of a house, builders must design the roof so that rainwater can flow easily into the gutters. A section of the roof can be represented on a coordinate grid. The roof slopes from point G(3, 6) to point H(9, 15).
 - i. Plot the points G(3, 6) and H(9, 15) on a coordinate plane and join them with a straight line to represent the roof.

- ii. Calculate the gradient of the roof.
 - iii. Discuss how the steepness of the roof affects the flow of rainwater during heavy rainfall.
 - iv. If the roof were less steep, what problems might occur with rainwater?
3. A road goes from point A(-10, 50) to point B (30, -110)? Find the gradient of the road.

Learning activity 5 : Graphing straight lines ($x = a, y = b, y = x$ and $y = mx + c$)



Activity 5.1: Graphing horizontal and vertical line

Instructions:

- Put learners in small groups
- Give each group coordinates that will make either horizontal straight lines or vertical straight lines.

- On graph paper, each group plots their assigned points and join them to form a straight line.
- Ask learners to state the observations made on the x and y values of the drawn lines.
- Ask learners to determine what equation this line would have.
- Ask learners Which line is the line $x=0$ and which line is the line $y=0$



Content tips:

- The ordered pairs where the x values are constant form a vertical straight line with equation $x = a$. where a is the constant value. (e.g $x = 2, x = 3, x = 0, x = -4 \dots$).
- The ordered pairs where the y values are constant form a horizontal straight line with equation $y = b$. where b is the constant value. (e.g $y = 1, y = 2, y = 0, y = -2 \dots$).
- The line $x = 0$ is the y – axis and the line $y = 0$ is the x – axis.



Activity 5.2 : “Same values ” ($y = x$)

Instructions:

- Put learners in small groups.
- Assign each group two coordinates with same value of x and y [e.g (1,1) and (4,4)]. Pairs should be in the first quadrant.

- join them to form a line.
- Learners to extend the line in both directions going passed the origin.
- Ask learners what they have observed about the x and y values on this line.
- Learners determine what equation could represent this line.



Content tips

- The matching coordinates indicate that the line passes through the origin (0,0) and rises diagonally.
- This means that the line being formed is the line $y=x$ where x and y are always equal.
- This line represents a 45° diagonal line on the Cartesian plane.



Activity 5.3: Opposing numbers ($y = -x$)

Instructions:

- Put learners in small groups.
- Assign each group two coordinates with opposing numbers like (2, -2) and (5,-5) in the fourth quadrant and join them.
- Learners to extend the line in both directions so that it passes through the origin.
- Ask learners what they have observed about x and y values on this line.

- Ask learners to determine the equation for this line.



Content tips

- The opposing coordinates form the line $y = -x$ which represents all points where the y-coordinate is the positive of the x-coordinate or vice versa.



Activity 5.4: Taxi Fare Investigation ($y = mx + c$)

Instructions:

- Learners analyse the scenario and answer the questions that follow.

Scenario: James pays a taxi fare made up of two parts: a fixed charge of K5 plus K3 for every kilometer he travels. The table below shows different distances travelled by James on three different days.

Distance travelled (km)	Total taxi fare(K)
1	
2	
3	

- i. Find the total taxi fare for each of the distances travelled and complete the table.
- ii. From the table, put the distance travelled and total taxi fare in ordered pairs.
- iii. Using x for distance travelled and y for total cost, write an equation that shows how the total taxi fare is calculated
- iv. Draw an appropriate cartesian plane.
- v. Plot the points and join them with a straight line where x-axis should represent distance travelled and y-axis total taxi fare.
- vi. Using the line drawn,
 - a. find the total cost when the distance covered is 5km, illustrate your work using doted lines on the graph.
 - b. Find the distance traveled when the total cost is K23



Content tips:

- For this particular situation the graph will not cross over to the third quadrant.
- The conventional equation of any straight line on the graph is $y = mx + c$.
- Where m is the gradient.

- c is the y -intercept (the point at which the line meets the y -axis and $x = 0$). In our scenario the constant value was the fixed charge.



Activity 5.5 ; Equations in real life situations

- Learners will answer the following questions individually
1. Bananas cost k2 each, and there is a packaging fee of K5. Write the equation for the total cost y in terms of the number of bananas x .
 - i. Draw the graph of the equation for the total cost.
 - ii. Use the graph to find the total cost of 10 bananas.
 2. Naomi starts with k50 talk time. She is charged K4 per minute. Write an equation for the remaining talk time y after x minutes.
 - i. Draw the graph of the equation for the remaining talk time.
 - ii. Using the graph find the number of minutes Naomi spoke when she remained with K38.
 3. A ramp goes up 2 meters high for every meter you move forward. Write the equation to show the height y in terms of the distance forward x .
 - i. Draw the graph of the equation of the height.
 - ii. Using the graph find the height of the ramp after 4 meters.

Learning activity 6: Finding the equation of a straight line given gradient and a point, gradient and intercept, double intercept form (two points).



Activity 6.1: Gradient and a point

Instructions:

- Learners answer the following questions in small groups
1. A bus company charge increases by K5 per km. when a customer has travelled 2km, the total cost is K18. Find the equation that represents the total cost y , for travelling x km. From the information given, give an ordered pair represents a point on the line, and what value represents the gradient of the line? Explain how you identified them.
 2. Water flows into a tank at 10 litres per minute. After 3 minutes the tank contains 50 litres. Find the equation representing the amount of water y , in the tank after x minutes. Identify one point that lies on the graph of the situation and the gradient of the line. What do each of these values represent in the context of the tank?
 3. A learner saves K15 per week. After 4 weeks, they have K90. Find the equation showing the total savings after x weeks. Which numbers in the problem form a point on the graph, and which number represents the rate of change (gradient)?
 4. A park ranger is marking a new straight hiking trail. The trail begins at a ranger station located at point B $(-1, 4)$. Due to the terrain, the trail must maintain a constant gradient of -2 . What is the equation of the line representing

this new hiking trail? Which information in the problem represents a point on the line, and which represents the gradient? How do these two pieces of information help determine the equation of the line.



Content Tips:

- In the situations given in activity 6.1 every rate of change is the gradient of the straight line representing the relationship between the two quantities.
- If the gradient (m) and a point (x_1, y_1) on the line are known, the equation of a straight can be found using:

$$y - y_1 = m(x - x_1)$$



Activity 6.2: Gradient and intercept

Instructions:

- Learners answer the following questions in small groups
 1. A school printing shop charges a fixed fee of K3 for using the printer and K0.50 per page printed. Let x represent the number of pages printed and y represent the total cost.
 - i. Determine the equation representing the cost.
 - ii. Identify the gradient and the y intercept for the equation connecting y and x .
 - iii. calculate the cost of printing 20 pages.

- iv. What can you observe about the value of m and c in the equation and the information given in the question.
- 2. A cyclist must move 4km to reach the start point of a race. From the starting point he rides at a speed of 10km per hour. Let x represent time in hours and y the total distance travelled.
 - i. Write the equation of the distance travelled.
 - ii. Determine the gradient m and the intercept.
 - iii. Determine the distance he covered after 3 hours of the race.



Content tips:

The equations formed in activity 6.2 are used to represent situations where quantities change at a constant rate.

If the gradient (m) and the y intercept $(0, y_1)$ on the line are known, the equation of a straight can be found substituting m and c directly into the equation $y = mx + c$.



Activity 6.3: Double intercept form

Instructions:

- Learners answer the following questions in small groups

1. A small truck can carry a maximum of 12 boxes of maize if it carries no bags of fertilizer. If it carries 6 bags of fertilizer, then it can carry no boxes of maize. Let x represent the number of boxes of maize and y the number of bags of fertilizer.
 - i. Draw a graph to illustrate this information on the cartesian plane.
 - ii. Determine the equation of the graph.
 - iii. state the x and y intercepts.
 - iv. write the equation of the line in the form $\frac{x}{a} + \frac{y}{b} = 1$.
 - v. compare the graph and the equation and make conclusions about a and b .



Content tips:

- Given the x and y intercept the equation of a straight line can be written in the form

$$\frac{x}{a} + \frac{y}{b} = 1$$

where:

- a is the x coordinate of the x -intercept (The point where the line crosses the x -axis), located at $(a,0)$
- b is the y coordinate of the y -intercept (The point where the line crosses the y -axis), located at $(0,b)$

Learning activity 7: Determining the gradient of a straight line from the equation



Activity 7.1: Discovering the Gradient from an Equation

Instructions:

- in small groups, learners analyse the scenario and answer the questions that follow

Scenario: A plane starts its journey from a station located 3 km east and 2 km north of a town center. As it travels, it gains altitude in a steady pattern. The path of the plane can be represented by the equation:

$$\frac{x}{2} - \frac{y}{4} = 1$$

Where:

- x represents the horizontal distance (in km east - west) from the town center
- y represents the altitude of the drone (in km)
 - i. Using the equation Identify the gradient of the plane's path from the equation.
 - ii. Explain what this gradient means in the context of the plane's movement.
 - iii. If the plane moves from $x = 2\text{km}$ to $x = 4\text{km}$, determine how much its altitude changes.

- iv. Due to bad weather plane follows a different path but still begins from the same point. The path it follows has the equation

$$y - 5x = -1$$

Use the equation to find the gradient.

- If the gradient was larger, how would the plane's path change?
- If the gradient was negative, what would this tell you about the plane's movement?
- Without changing the gradient, how could the equation be changed so that the plane starts higher in the air?



Content tips:

- From activity 7.1 the equations need to be written in standard form $y = mx + c$ and then learners extract the gradient from the equation. The gradient is the coefficient of x .

Learning activity 8: Drawing parallel lines and comparing their gradients



Activity 8.1: Parallel lines

Instructions:

- In small groups, learners analyse the two scenarios and answer the questions that follow ;

Scenario 1: A road rises steadily as it moves away from a town. At the town center the road is 2 m above sea level, and for every 1 km travelled east, the height increases by 3 m. Let

- (x) = distance from the town (km)
- (y) = height above sea level (m)

Scenario 2: A water tank already contains 5 litres of water. Water flows into the tank at a constant rate of 3 litres per minute.

- x = time in minutes
- y = amount of water in the tank (litres)

- i. Determine the equations for scenario one and two.

- ii. What is the gradient of each equation?
- iii. What do you observe about the two gradients?
- iv. What does the gradient represent in each situation?
- v. On the same cartesian plane, draw the graph for each of the equations in (i).
- vi. What conclusions can you make from the graph of the two lines and their gradients.



Content tips:

- In activity 8.1 the gradients for both lines are the same (the gradient for both lines is 3), hence making the two lines are parallel.
- If the gradients are different the lines will have different slopes and will eventually intersect.
- Given that two lines P and Q are parallel then gradient of p m and gradient of Q m are equal. Therefore

Learning activity 9: Drawing perpendicular lines and comparing their gradients



Activity 9.1: Solar Panel Frames in a Community Park

Instructions:

- Learners analyse the scenario and answer the following questions in small groups

Scenario: In a community park in Lusaka, engineers install a long metal support beam for solar panels. For every 4 meters you move east, the beam rises 2 meters.

Let:

- x = horizontal distance (meters)
- y = height (meters)

Later, a second beam is installed to support the structure, and it must cross the first beam at a right angle for stability. For every 2 meters you move east, this second beam drops 4 meters.

- i. Determine the gradient of the of the two beams.
- ii. Multiply the two gradients and state what you can observe.
- iii. If both line representations pass through the origin
 - a. Write the equations of the two beams
 - b. Draw both lines on the cartesian plane
- iv. Measure the angle at the point of intersection
- v. Why does one slope have to be negative?
- vi. What happens if both slopes are positive?
- vii. Can two positive gradients ever form a right angle?
- viii. What pattern do you see between the gradients of perpendicular lines?



Content tips:

- Learners should notice that the gradients are **negative reciprocals** and their **product is -1** .) for instance in activity 9.1 , the two gradients are 2 and $-\frac{1}{2}$ and $2 \times \left(-\frac{1}{2}\right) = -1$



Assessment

1. Draw a rectangular park on the cartesian plane and state the coordinates of its vertices.
2. A football pitch is represented on a Cartesian plane with corners at A(0,0), B(0,90), C(120,90), and D(120,0).
 - a) Plot the pitch.
 - b) Find the coordinates of the centre of the football pitch.
3. A road runs through two towns represented by points M(2,-3) and N(6,5).
 - a) Determine the gradient of the road.
 - b) Write the equation of the line passing through M and N.
4. A line has the equation $3x + 2y = 12$
 - i. Determine its gradient and y-intercept.
 - ii. Draw the graph on the cartesian plane.

5. A road is represented by the line $y = -13x + 4$. Another road is given by $y = -13x - 2$.
- a) Are the two roads parallel?
 - b) Suggest a real-life implication of parallel roads in city planning.
6. A farmer draws two paths across his land: one represented by the line through points $P(0,0)$ and $Q(2,4)$, and another through points $R(0,0)$ and $S(4,-2)$.
- a) Find the gradients of both paths.
 - b) Determine if the paths are perpendicular.
7. A city planner models two roads with equations:
- Road 1: $y = 3x + 2$
- Road 2: $y - 3x = -5$
- a) Identify the gradients of both roads.
 - b) Conclude whether the roads are parallel and give reasons.

Expected Standard: Straight lines used correctly to solve problems in real life situations.

TOPIC 2: RELATIONS AND FUNCTIONS

INTRODUCTION

Relations and functions are important concepts in Mathematics that help us understand how different quantities are connected. A **relation** describes a relationship between two sets of values, showing how elements from one set correspond to elements in another. A **function** is a special type of relation where each input is linked to exactly one output. Relations and functions are widely used in areas such as science, economics, engineering, and technology, helping us understand how one quantity changes in response to another.



Key terms: Relation, Function, Domain, Range, Ordered Pair, Mapping (Arrow Diagram), Input, Output, Independent Variable, Dependent Variable, One-to-One Relation, Many-to-One Relation, One-to-Many Relation, Many-to-Many Relation

General competence(s):

Analytical Thinking, Problem Solving, Creativity, Critical Thinking, Collaboration, Financial Literacy, Digital Literacy



In everyday life, some quantities depend on others, such as money earned depending on hours worked. How can mathematics help us represent and understand these relationships?

Sub-Topic 1: Relations

Introduction

Relations help us describe and understand connections between different quantities or sets. In Mathematics, relations can be represented in various ways such as ordered pairs, tables, arrow diagrams (mappings), and graphs. These representations allow us to analyse how elements in one set are linked to elements in another set and help us determine different types of relations such as one-to-one, many-to-one, one-to-many, and many-to-many.

Specific competence: Apply relations to model and solve problems

Learning Activity 1: Exploring the types of relations in real life situations (one to one, one to many, many to one, many to many)



Activity 1.1: Exploring relations

Instructions

- Divide learners into small groups.
- Provide each group with at least **4 cards**. Each card should contain a **pair of real-life sets**, for example:
 - Learners and ID numbers
 - Teachers and subjects taught
 - Learners and their favorite sports
 - Countries and languages spoken

- Ask learners to examine each pair of sets and discuss **how the elements in the first set relate to the elements in the second set.**
- Ensure that the cards given to each group include examples representing **all four types of relations:**
 - one-to-one
 - one-to-many
 - many-to-one
 - many-to-many



Discussion

- Each group to discuss the following questions
 - i. For each pair given, identify what kind of things/ elements are in the first set and the second set.
 - ii. What is the connection between items in the first set and items in the second set?
 - iii. Find the pair where each item in the first set is linked to only one item in the second set, and each item in the second set is also linked to only one item in the first set. Justify your answer.
 - iv. Find the pair where one item in the first set is connected to more than one item in the second set, while each item in the second set is connected to only one item in the first set. Justify your answer
 - v. Find the pair in which several items in the first set are connected to the same single item in the second set. Justify your answer

vi. Find the pair in which several items in the first set are connected to several items in the second set?

Justify your answer.

- Groups present their solutions and justifications to the entire class.
- After the presentations, in their groups, learners to assign the conventional names (one to one, one to many, many to one, many to many) to each of the relations described.



Content tips:

- The connection between the elements of the two sets is known as the **relation** between them.
- A relation describes how elements in one set are connected or matched to elements in another set.
- When each item in the first set is connected to only one item in the second set, and each item in the second set is also connected to only one item in the first set, the pair shows a **one-to-one relationship**.
- When one item in the first set is connected to more than one item in the second set, while each item in the second set is connected to only one item in the first set, the pair shows a **one-to-many relationship**.
- When several items in the first set are connected to the same item in the second set, the pair shows a **many-to-one relationship**.
- When several items in the first set are connected to several items in the second set, the pair shows a **many-to-many relationship**.

Learning Activity 2: Representing relations using arrow diagrams/ mappings, ordered pairs



Activity 2.1 Representations of relations

Instructions

- In the same groups formed in activity 1.1, learners list at least 5 elements in each of the sets given.
- Learners to then represent their relations diagrammatically, clearly depicting the relation and present to the class and justify their choice.

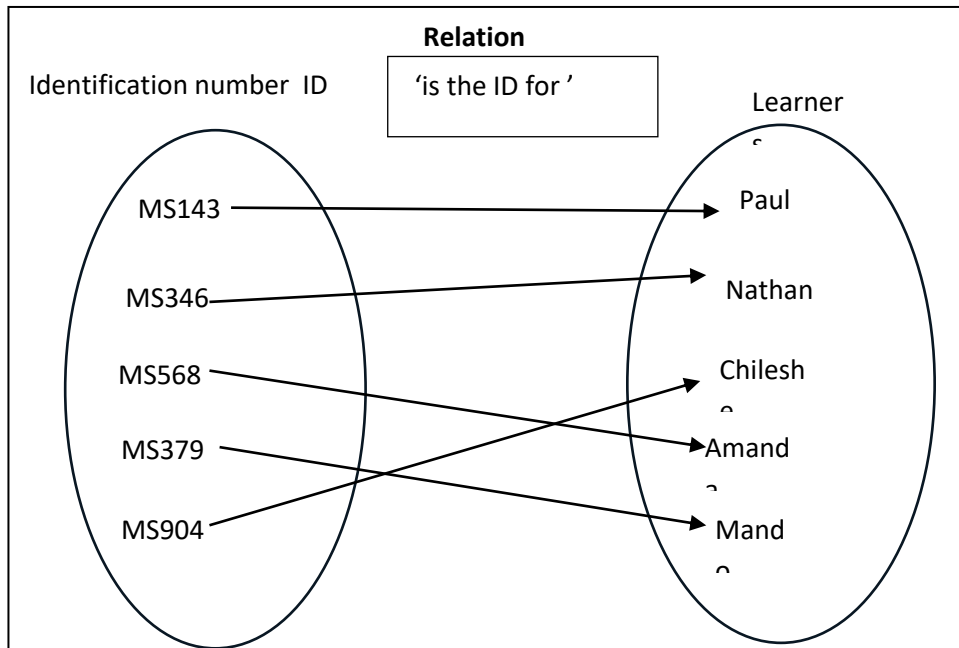


Content tips:

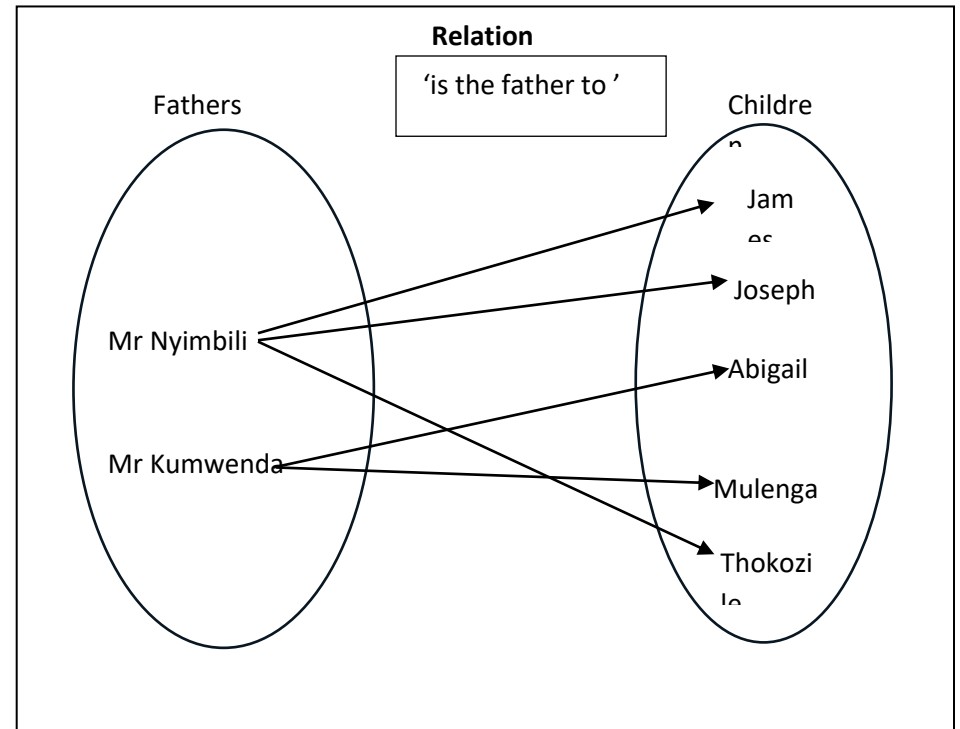
- The conventional way of diagrammatically representing a relation is through an arrow diagram.
- An Arrow diagram provides diagrammatic representation of a relation, as shown in the following diagrams:

Draw more arrow diagrams

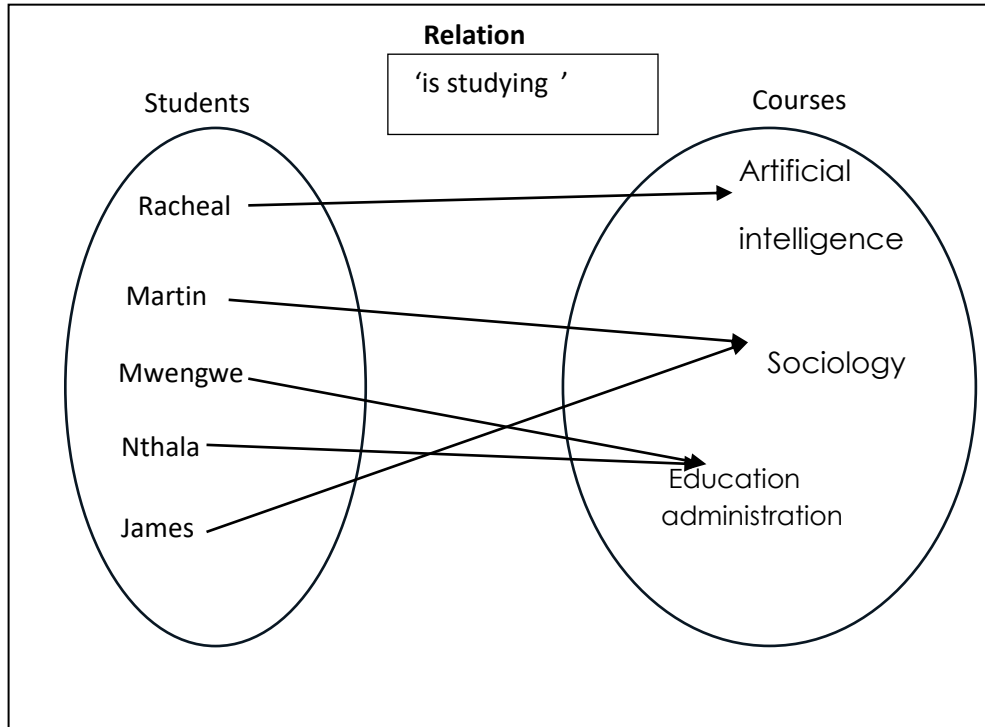
i. One to One



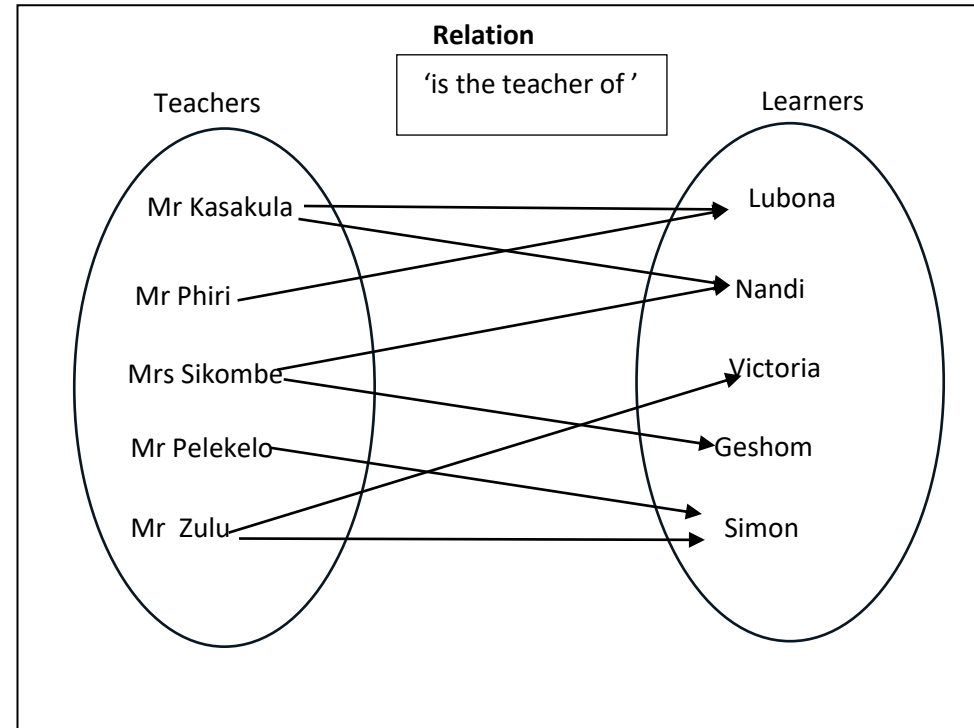
ii. One to many



iii. Many to one



iv. Many to many



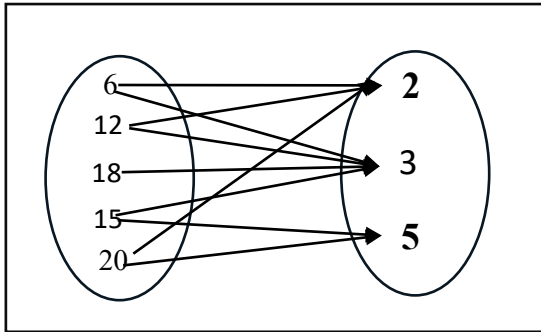


Activity 2.2: Pair work (match the elements)

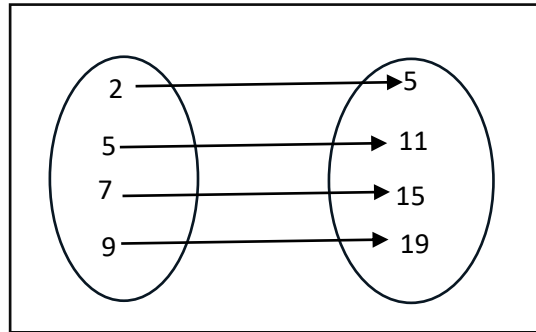
In pairs learners answer the following questions.

1. State the relation and type of relation connecting set A to set B in each of the following and present them as ordered pairs.

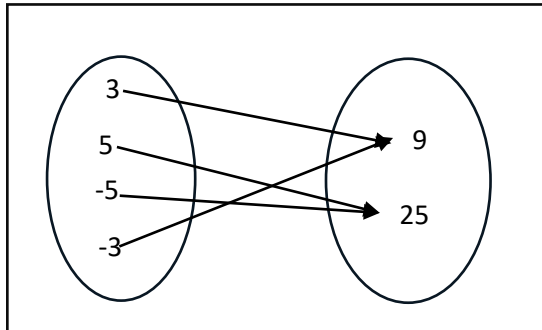
a)



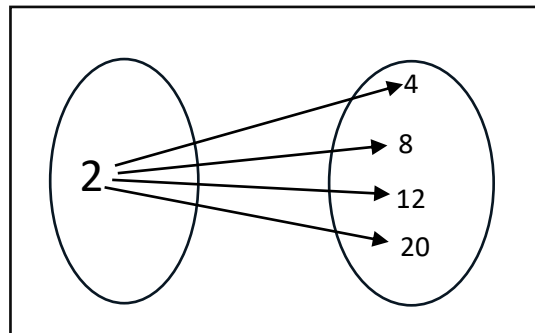
b)



c)



d)

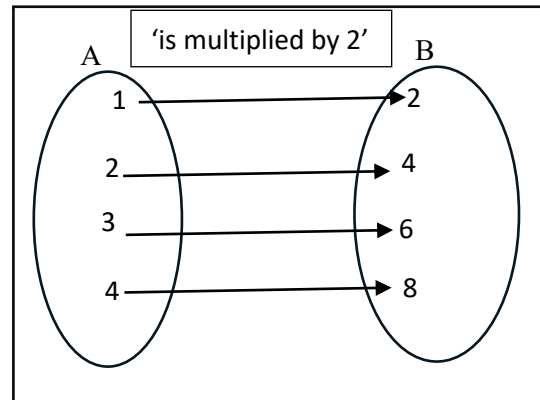


2. For each of the following, set A and the relation between it and set B has been given. Find the elements of set B and represent this relation using ordered pairs and on an arrow diagram.
- a. Set $A = \{1, 2, 3, 4\}$, Relation: “**is multiplied by 2**”
 - b. Set $A = \{1, 2, 3, 4\}$, Relation: “**add 3**”
 - c. Set $A = \{2, 3, 4, 5\}$, Relation: “**is a square root of**”
 - d. Set $A = \{1, 2, 3\}$, Relation: “**is a multiple of**”
3. Which representation (ordered pairs or arrow diagrams) helped you understand the relation better? Explain why.



Content tips:

- **One-to-one:** only one arrow from each element in set A to exactly one element in set B.
- **Many-to-one:** more than one arrow from different elements in set A pointing to the same element in set B.
- **One-to-many:** more than one arrow from a single element in set A pointing to different elements in set B.
- **Many-to-many:** more than one arrow from several elements in set A pointing to several elements in set B.
- **From activity 2.2,** in question;
 - 1. a), the relation is ‘is a multiple of’, this is a many to many relation.
 - 2. a) , Set $B = \{2, 4, 6, 8\}$, ordered pairs $\{(1,2), (2,4), (3,6), (4,8)\}$ and arrow diagram :



Expected Standard: Relations to model and solve problems applied correctly

Sub-Topic 2: Functions

Introduction

Functions are a special type of relation that describe a clear connection between two quantities. In a relation, elements can be connected in different ways, but in a function each element in the first set is connected to only one element in the second set. Studying functions helps us understand how one quantity depends on another, identify patterns, make predictions, and solve real-life problems.

Specific competence: Apply Functions to model and solve problems

Learning Activity 1: Determining which relations are functions and non – functions



Activity 1.1: Path prediction

Instructions:

- In small groups learners analyse the following questions

Scenario:

Two friends, Chanda and Tandi, walk from home to school every day. Chanda always uses the same road from her home straight to the school because there is only one path she can take. Tandi sometimes uses

different roads. Some days she goes through the football field, other days she passes by the clinic, and sometimes she goes through the shops.

- i. If Chanda leaves home, which path will she take?
 - ii. If Tandi leaves home, can we be sure which path she will take? Give reasons for your answer?
 - iii. Whose movement is easier to predict?
 - iv. Why is it easier to know where Chanda will go than Tandi?
- Groups present their work
 - After consolidation, learners return to their groups and answer the following
'A function shows how one quantity determines another. If we know the input, we should be able to predict exactly one output. Looking at the four relations (one to one, one to many, many to one, many to many), identify which relations follow this rule and is therefore a function'.



Activity 1.2: Functions and non-functions

Instructions

- **In pairs learners answer the following questions**
 - i. Given that a relation is a function if each value of set A (input) is paired with exactly one value of set B (output). Determine which of the following are functions and which ones are not and justify your answer.

1. $(1,2), (2,4), (3,6), (4,8), (5,10), (6,12)$
 2. $(1,2), (1,4), (2,5), (3,6), (4,7), (5,8)$
 3. $(1,2), (1,3), (2,3), (3,4), (4,2), (5,4)$
 4. $(1,5), (2,5), (3,7), (4,7), (5,9), (6,9)$
 5. $(10,1), (20,2), (30,3), (40,4), (50,5), (60,6)$
 6. $(1,100), (2,100), (3,200), (4,200), (5,300), (6,300)$
- ii. Plot each set of ordered pairs on a separate cartesian plane, connect all the points.
 - iii. How can we use a vertical line to determine which one is a function and which one is not a function.



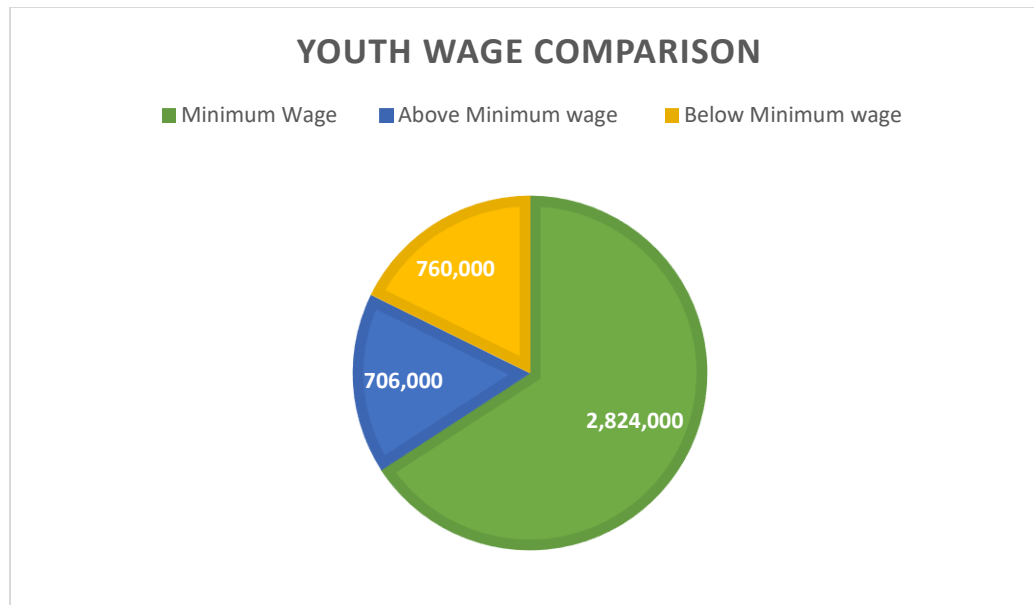
Activity 1.3: Pair work / small groups

- In pairs/small groups learners analyse and solve the following problem posed.

Scenario:

The government of Zambia requires that workers receive at least a minimum hourly wage. A survey was conducted among youths in different towns.

The pie chart shows the number of working youths according to how their wages compare with the minimum wage.



- What percentage of working youths earn more than the minimum wage?
- Write a formula for converting the numbers from the pie chart into percentages.
- Does your formula from part (b) represent a function? Give reasons for your answer.
- Suppose the minimum wage in Zambia is K25 per hour. Write an equation for earnings as an equation of the number of hours worked. Let h represent the number of hours worked and k the total wage.
- Is the equation in (d) a function or not. Give justification for your answer.

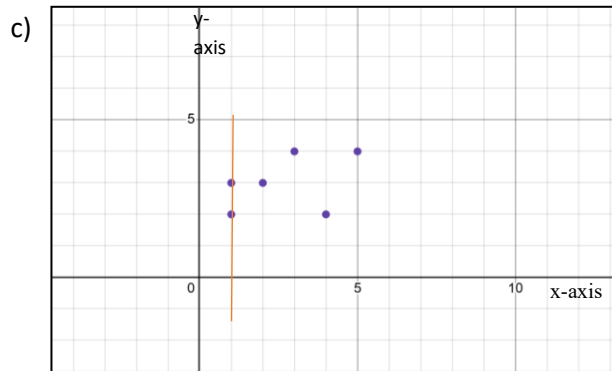


Content tips:

- A Function is a rule that relates an input to a unique output. It acts as a machine mapping each element in the first set to exactly one element in the second set.
- If a vertical line touches the graph at more than one point as it traverses from left to right then relation is not a function.

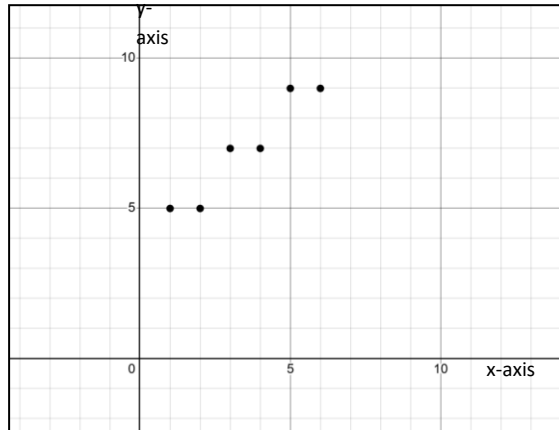
For instance :

- In activity 1.2 part c) the vertical line intersects the graph at more than one point (at $x = 1$), the graph fails the test, hence it is not a function



- For part d) the vertical line **intersects the graph at only one point**, the graph **passes the test**, hence it **is a function**.

d)



- Online graphing tools such as desmos can be used to give a visual demonstration of the vertical lines movement from left to right. Click on the link to viewpart f)

<https://www.desmos.com/calculator/pzsv6hh2f4>



Activity 1.3: Function or not a function

- The following tables show ordered pairs. Determine which tables represent functions and which ones are not. Give reasons for your answer.

a)

Input	output
13	60
13	63
14	64
14	68
16	68

b)

Input	Output
5	5
8	11
13	21
16	27
21	37

c)

Input	Output
-3	9
0	0
3	9

d)

Input	Output
1	2
1	3
2	2

4	16
-4	16

3	3
---	---

Learning Activity 2: Representing a function as $f(x) = a, f: x \rightarrow y$



Activity 2.1: The Machine

Instructions

- Give the learners to work on these questions in pairs or small groups (3-5)

Scenario:

A motorcycle rental shop charges K12 to rent a bike plus K4 for every hour used. The owner decides to use a machine to help him determine the cost given the number of hours the bike has been taken. The machine takes the number of hours (input), performs a rule, and gives a new number (output). The output is the total cost of the rental. If the machine performs the rule 'multiply by 12 then add 4' the input and outputs are given in the following table.

Input	Output
1	16
2	28
3	40
4	52

- i. Write this function as an equation where y represents the output and x represents the input, ensure that y is the subject.
 - ii. This machine (function) acts on x to give y and if we call this function f , then f is a function of x and can be written as $f(x)$. An equation of a line becomes a function when we represent the output as $f(x)$ instead of y . Rewrite the equation in i) as a function.
 - a. What value does $f(4)$ produce
 - b. What value does $f(10)$ produce
 - c. What value of x would produce 100
- Learners to present their solutions to the class for consolidation.



Activity 2.2 The Machine (continued)

Given another function machine that takes a number (input) performs the rule 'multiply by 6 then subtract 2' the input and outputs are given in the following table:

Input	Output
1	4
3	16
5	28
7	40

- Write this function as an equation where y represents the output and x represents the input.
- Denote this function with a letter of your choice and use that letter to rewrite this equation as a function of x .
- With the letter you have chosen use function notation ($f(x) = a$) to show that when the input is 3 the output is 16.



Content tips:

- A function is denoted by $f(x)$ and is read as “f of x”, where
 - o f is the name of the function (the rule or machine).
 - o x is the input.
 - o $f(x)$ is the output when x goes through the rule.
- For instance, in activity 2.2 the function notation $f(2) = 10$ represents;
 - o “f of 2 equals 10.”
 - o “When the input is 2, the value of the function is 10.”
 - o The value of x is substituted in the equation to get the value of y .

$$y = 6x - 2$$

$$y = 6(2) - 2$$

$$y = 12 - 2$$

$$y = 10$$



Activity 2.3: The machine reversed

- In pairs the learners answer the following questions.
 - a. Given that the equation connecting x and y is $2x = 5 - y$. Rewrite this using function notation $f(x)$.
 - b. $f: x \rightarrow y$ is another way of denoting a function where $(:)$ is read as '**such that**' and the arrow $\rightarrow$ means '**maps to**' or '**changes into**'. Rewrite the function in a) using the notation $f: x \rightarrow y$.
- With the teacher's guidance, learners present their solutions.



Content tips:

- The function notation $f(x)$ is used when y is the subject of the formula. For instance, in activity 2.3, the equation $2x = 5 - y$, will be rewritten as $y = 5 - 2x$.
- The equation $2x = 5 - y$ can be written in function notation as $f(x) = 5 - 2x$ and as $f: x \rightarrow 5 - 2x$.



Activity 2.4: Function representation

- Learners to answer these questions individually

Task : For each of the situations given find a function that can determine the output given a particular input and determine the input needed to produce a given output. Write this function in the form $f(x)$ and $(f: x \rightarrow y)$

- i. A young tree is 50 cm tall and grows 3 cm each month. (the function should predict the growth of the tree after a certain number of months)
- ii. A home uses 2 units of electricity every hour. (the function should predict the amount of electricity used after a certain number of hours)
- iii. A worker earns K50 per day. (the function should predict the earnings after a certain number of days)

Learning Activity 3: Finding the range given the domain and vice versa



Activity 3.1: Range and Domain adventure

Instructions

- In groups, learners analyse the scenario and answer the following questions:

Given that the set of inputs is called the domain and the set of outputs is called the range analyse the following scenario.

In a town, a taxi company charges passengers according to the distance they travel. There is a fixed starting charge of K10 when you enter the taxi. The taxi then charges K5 for every kilometer travelled. The fare can be calculated using the function $y = 5x + 10$, where (x) is the distance travelled in (km) and (y) is the total fare (Kwacha).

- i. A driver records the following distances travelled by passengers: distance in km {1 km, 3km, 5km, 8km}
 - a. Calculate the total fare for each trip.
 - b. Present the ordered pairs for each trip in a table, the header of the table should clearly show the Domain and the range.
- ii. Another driver only remembers the fares collected that day: fare in Kwacha {K15, K25, K40}
 - a. Work backwards to determine the distance travelled for each fare.
 - b. Write the domain.
 - c. Present the ordered pairs for each trip in a table, the header of the table should clearly show the Domain and the range.
 - d. Could two passengers travel different distances but pay the same fare in this situation? Explain your reasoning.



Activity 3.2: Finding the Range and Domain

- **instructions**

Learners answer the following questions individually.

1. Mobile Phone Airtime

A mobile network charges a **connection fee of K2** for making a call and **K1.50 for every minute** you talk. The cost of the call is given by the function $f(x) = 1.5x + 2$, where (x) is the number of minutes and $f(x)$ the total cost in Kwacha.

- i. **Tobela** makes calls on 5 consecutive days and the minutes she spends are given as {1min, 3min, 5min, 10min, 12min. Calculate the **cost of each call** and write the range. Present these ordered pairs in a table.
- ii. **Sheila** has chosen to take calls on credit and only pay at the end of the week. She receives a total bill of K19.50. Sheila cannot believe she spent so much money and so she decides to complain to mobile network provider. The network provider sends her this fee breakdown for cost of each call she made in the week {K3.5, K6.5, K 9.5}. Help Sheila confirm whether the number of minutes the network provider charged her for is the same as the one she has on her call record, if her call record is showing a total of 9 minutes in that week.

2. School Printing Service

A school printing shop charges K3 to start a print job and K2 for every page printed. The shop uses the function $f(x) = 2x + 3$ to predict total cost. Where (x) is the number of pages and $f(x)$ is the total cost

Ethel goes to the print shop on four consecutive days and prints the number of pages given in the set where each element presents the number of pages printed on a particular day {2, 4, 6, 10}.

- a. Find her expenditure for each of the days. Present the costs as a set.
- b. Present these findings in a table, clearly indicating the column of Domain and the range.
- c. Find Ethel's total expenditure in that week.

Learning Activity 4: Finding a function given a set of ordered pairs



Activity 4.1: Mystery Machine

Instructions

- In pairs learners analyse the scenario and provide solutions to the problems posed.

a. Juice shop

A small juice shop keeps track of how much customers pay depending on how many bottles of juice they buy. The owner records the following information:

Number of bottles (x)	Total cost in Kwacha (y)
1	7
2	10
3	13
4	16

The shop is purchased by a new owner who does not know the connection between the number of bottles and the total cost but just has the information in the table. His wish is to find out the function being used to determine the total cost. To help the new owner:

- i. Present the information in the table as ordered pairs
- ii. Use the ordered pairs to determine the function of the total cost given a number of bottles.
- iii. The new owner decides to increase either the constant charge or the cost per bottle. Which of the two should the new owner increase to increase profit but also not discourage customers?

b. Vitumbuwa sales

At a school in Mulobezi, Mrs Tembo sales tasty fried snacks called vitumbuwa. She charges customers depending on the number of vintumbuwas they buy and adds a fixed charge for packaging.

Margret wants to take over Mrs Tembo's business. Mrs Tembo then decides to record some of her sales for the day to help Margret have an idea of her sales. The sales information is given in the following table.

Number of Vitumbuwa (x)	Total cost (Kwacha) (y)
1	9
2	12

Number of Vitumbuwa (x)	Total cost (Kwacha) (y)
3	15
4	18

- Help Margret determine the function of the total cost given the number of vitumbuwa.
- Margret sales to 6 customers as follows: 3 Vitumbuwas, 8 Vitumbuwas, 5 Vitumbuwas, 10 Vitumbuwas, 12 Vitumbuwas, 9 Vitumbuwas. Determine how much she charged each customer.
- Present Margrets sales and total cost as ordered pairs.



Content tips :

- The function can be found by getting at least two ordered pairs and finding the equation ($y = mx + c$).
With this equation the learners can describe the function using $f(x)$.
- In activity 4.1;
 - If the new owner wants to increase profit without discouraging customers, it is better to increase the constant charge slightly rather than the cost per bottle.

- The cost per bottle (3) affects every bottle purchased
- The constant charge (4) is paid once regardless of how many bottles are bought.

Learning Activity 5: Investigating the inverse of a function given using real life situations (*order of parking and unparking a box ...*)



Activity 5.1: Parking and unparking a box

A warehouse worker parks storage boxes in numbered spaces. To keep track of them, the worker uses a rule to assign each box number to a parking space number.

Parking rule:

To find the parking space for a box, the worker uses the rule: Multiply the box number by 2 and then add 1.

Box Number (Input)	Parking Space (Output)
1	3

Box Number (Input)	Parking Space (Output)
2	5
3	7
4	9

Task

- i. Write the function rule that connects the box number x to the parking space y .
- ii. Now the worker wants to unpark the boxes.
He only knows the parking space number and needs to determine the box number.
 - a. If a box is in parking space 11, which box number is it?
 - b. If a box is in parking space 15, which box number is it?
- iii. Describe the rule used to go backwards from parking space to box number.
- iv. What operations were used in the parking rule?

- v. What operations must be done to reverse the process?
- vi. In what order should these reverse operations be applied?
- vii. Why must the operations be reversed in the opposite order?



Content tips:

- In activity 5.2 the parking rule is $f(x) = 2x + 1$, then the inverse rule (unparking rule) reverses the steps:

First step: Subtract 1

Second step: Divide by 2

- Therefore, the reverse rule is $f'(x) = \frac{x-1}{2}$.
- Changing the subject of the formula for the equation from y to x can aid in finding the inverse. For instance, in this activity the function $f(x) = 2x + 1$ gives the equation

$$y = 2x + 1$$

- Making x the subject gives;

$$x = \frac{y - 1}{2}$$

This then becomes the rule/function for the reverse action. The set of outputs then becomes the set of inputs, and the set of inputs becomes the set of outputs. Therefore, x becomes y and y becomes x .

$$y = \frac{x - 1}{2}$$

The reverse function (inverse function) then becomes

$$f'(x) = \frac{x - 1}{2}$$

Learning Activity 6: Determining a function without an inverse



Activity 6.1: 'can we reverse the rule'

Instruction

- In pairs or small groups (3 to 4) learners analyse the scenario and answer the questions that follow.

Scenario:

A factory makes square tiles for flooring. Some tiles are labeled with positive measurements, and some are labeled with negative measurements to represent direction on a design grid. However, when calculating the area, the factory squares the side length.

The rule used is:

$$\text{Area} = (\text{side length})^2$$

The ordered pairs are given in the following table

Side Length $x(\text{cm})$	Area $y (\text{cm}^2)$
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

- i. Given that a function's inverse follows the same rules of a function, determine if this function has an inverse? Justify your answer.



Content tips:

- If two inputs produce the same output, we cannot reverse the rule uniquely.
- From activity 6.1 moving in the reverse order will give a one to many relation. This is a type of relation that is not a function.

Learning Activity 7: Exploring Composite Functions



Activity 7.1: “Two Machines Working Together”

Instructions

- Learners analyse the scenario and answer the following questions.

Scenario : In a factory there are two machines that are used to generate production costs, one after the other. Each machine changes a number according to a rule.

Machine A:

‘Multiply the number by 2’

$$A(x) = 2x$$

Machine B:

‘Add 3 to the number’

$$B(x) = x + 3$$

Input	Machine A	Machine B
1	2	5
2		
4		
6		

- Complete the table with the appropriate outputs.
- How can the two functions be written as one?
- Use the new function to determine the output when the input is 8.
- If the input was placed in machine B first and then machine A, complete the following table

Input	Machine B	Machine A
-------	-----------	-----------

Input	Machine B	Machine A
1	4	8
2		
4		
6		

- v. How can the two functions be written as one?
- vi. Use the new function to determine the output when the input is 8.
- vii. Compare the results in iii and vi, is the result the same? What conclusions can you make about the composite functions?



Content tips:

- If we have two functions f and g , the application of the two functions one after the other can be expressed as $f[g(x)]$.

- $f[g(x)]$ is the composition of two functions and it is read 'f of g of x' and it is referred to as a composite function.
- Composite functions can also be denoted as $(f \circ g)(x) = f[g(x)]$.
- In a composite function the second function uses the output of the first function as its input.
- The second function replaces the x value with first function.
- For instance

First function

$A(x) = 2x$ and the second function $B(x) = x + 3$, every value of x in B(x) will be replaced with A(x), giving

$$B(Ax) = (2x) + 3$$

When first function is $B(x) = x + 3$ and the second function $A(x) = 2x$, every value of x in A(x) will be replaced with B(x) giving

$$A(Bx) = 2(x + 3)$$

- The order in which the function is applied is important.
- When the first function is A and the second function is B the composite function will be $B[A(x)]$ or $(B \circ A)(x)$. Here, A(x) replaces the value of x in the function .
- When the first function is B and the second function is A the composite function will be $A[B(x)]$ or $(A \circ B)(x)$.

Learning Activity 8 : Solving problems involving Composite Functions



Activity 8.1: Two Machines in one

1. Scenario: A customer wants to buy a pair of shoes from a shop. The market price of the shoes is K200.
Before the customer pays, the shop applies two charges:
 - First charge; The shop first adds 16% of the original price of the shoes and K5 fixed charge.
 - Second charge; the shop then adds a delivery fee of K20.
 - i. Write two functions one for each of the charges.
 - ii. Present these functions as composite function.
 - iii. Using the composite function determine the cost of the shoes.
 - iv. Would the final price be the same if the shop added the delivery fee first? Explain your reasoning.
2. Scenario: A delivery company uses a plane to deliver packages in a city. The plane's movement is controlled by two rules:
 - The distance x (in km) that the plane flies determines the time t (in minutes) it takes:
$$t = f(x) = 4x$$
 - The time t the plane spends flying determines how far north y (in km) it moves:

$$y = g(t) = 2t + 1$$

The city map is represented on a coordinate plane, where:

- x represents east–west distance and y represents north–south distance.
- i. Find the composite function that gives the north–south position y directly from the east–west distance x . (Find $g(f(x))$.)
- ii. If the plane travels 3 km east, determine its north–south position.
- iii. Write the ordered pair (x, y) representing the plane's position after traveling 3 km east.
- iv. Plot the point on a coordinate plane and explain what the coordinates represent in the real situation.
- v. If the plane travels 1 km, 2 km, 3 km, and 4 km east, calculate the corresponding y -values using the composite function and plot the points.
- vi. What kind of geometric pattern do the plotted points form?



Content tips:

- Composite functions represent a chain of dependent processes; output of one process becomes input of the next process.
- Functions and coordinate geometry complement each other to give learners a deeper understanding of a concept. In activity 8.1 the geometric pattern made is a straight line.



Assignment: Exploring Relations and Linear Functions in Real Life

In many everyday situations, one quantity changes depending on another. Your task is to investigate a real-life situation in your everyday life and represent the relationship using different mathematical forms and determine whether the relationship forms a function.

Task Instructions

1. Choose a Real-Life Situation: Select one situation where one quantity depends on another.
2. Represent the Relation (function): Create at least five ordered pairs showing how the two quantities are related. For instance, input (x) and output (y).
3. Justify why you believe this Relation is a Function.
4. Represent the Relation in Different Ways: Table of values, Equation of the linear function, Graph on the Cartesian plane. Your equation should be written in the form: $y = mx + c$.
5. Using your graph:
 - i. Identify the gradient of the line.
 - ii. Identify the y-intercept.
 - iii. Explain what these values mean in the real-life context you selected.
6. Use your equation to predict values not shown in your table.



Content tips:

- Learners should be graded based on the authenticity of their own real life situation and accurate connection of their situation.

TERM TWO (2)

TOPIC3 : ALGEBRA

Introduction

In Form 1, learners were introduced to the basics of algebra, such as variables, constants, coefficients, and terms, and they practiced simplifying expressions. To build on this foundation, in form 2, learners will now focus on the basic processes of algebra.



Key terms: Monomial, Binomial, Linear expression, Quadratic expression, Simplification, Factorization

General competences: Analytical thinking, Critical thinking, Problem-solving, Communication, Collaboration, Digital literacy



Key question



Can we rewrite mathematical expressions in different but equivalent forms to make them easier to work with?

Sub-Topic: Basic Processes of Algebra:

Introduction

The basic processes of algebra include expanding, simplifying and factorizing algebraic expressions. These processes form the building blocks for higher mathematical concepts and practical applications in science, technology, and everyday life.

Specific competence: Use algebraic expressions in problem-solving

Learning activity 1. Exploring real life situations involving expanding algebraic expressions (monomial by binomial and binomial by binomial)



Activity 1.1 Buying school supplies

Instructions:

Put learners in small groups and let them do the following activity:

Scenario: Chanda goes to a shop to buy school exercise books and pens. Each exercise book costs k8, but the learner does not know how many exercise books she will buy. Each pen costs k3, and she buys 4 pens.

- Let x represent the number of exercise books bought, write an expression for the total cost of the exercise books and pens.
- Given that the total cost of the books and pens can be written as $4(2x + 3)$, compare the expression in (a) with this one. What steps would we take to convert the expression in (b) to the expression in (a)



Content Tips:

- To expand the product of a single term (monomial) by a two term expression (binomial), multiply each term in the brackets by the single term. For instance, In activity 1.1,

$$4(2x + 3)$$

$$4(2x) + 4(3)$$

$$8x + 12$$



Activity 1.2 Expansion exploration

Learners answer the following questions individually

1. A builder in Kabwe purchases $5x$ cement blocks. Each block costs $(y + 10)$ kwacha, where y is the market price and 10 kwacha is the delivery fee per block. Write an expression for the total cost of buying and transporting the cement blocks.
2. A group of students hires $(2x)$ taxis to travel to Lusaka. Each taxi charges $(y + 50)$ kwacha, where y is the base fare and 50 kwacha is the fuel surcharge. Determine the total cost of hiring the taxis.



Activity 1.3: Expansion exploration

Instructions:

- Put learners in small groups
- Provide work cards to each group with two binomial expressions (length and width of rectangular garden) and the expanded form (area) written on them.
- Ensure that both terms in each binomial are different variables

For example:

Garden length: $(x + m)$ meters

Garden width: $(x + n)$ metres

- Ask the learners to determine the area of the rectangular garden using the given dimensions
Then give them work cards with the expanded form of the expressions given, depending on the variables in the expression given. For instance, expanded expression for the area: $x^2 + mx + nx + mn$.
- Learners to then compare your area with the expanded form given. What have you observed about the two forms?



Content tips:

- When multiplying a binomial by a binomial, each term in the first binomial is multiplied by each term in the second binomial.
- The FOIL method can be used to expand binomial by binomial, where

F - multiply the **F**irst terms $x \times x = x^2$

O - multiply the **O**uter terms $x \times n = xn$

I - multiply the **I**nnner terms $n \times x = nx$

L - multiply the **L**ast terms $n \times n = n^2$

- For instance: in activity 1.1 $(x + n)(x + n)$



Activity 1.4: Expansion exploration

Instructions

- Learners to answer the following questions individually:
1. A shopkeeper in Lusaka sells $(x + 50)$ shirts, where (x) is the number of shirts ordered by schools and 50 extra pieces are kept in stock. Each shirt costs $(y + 7)$ kwacha, where (y) is the market price and 7 kwacha is the tailoring fee. Write and simplify the expression for the total cost.
 2. An event organizer rents $(2x + 10)$ chairs for a conference. Each chair costs $(y + 5)$ kwacha, where (y) is the rental fee and 5 kwacha is the transport charge. Write and simplify the expression for the total cost.
 3. A farmer in Chipata has a rectangular vegetable garden. Its length is $(x + 5)$ meters and width is $(x + 3)$ meters. Determine the area of the farmer's vegetable garden.

**Learning activity 2: Model life situations using algebraic expressions and simplify them
(managing budgets, planning, production, tracking expenses, scheduling activities,...)**



Activity 2.1 Modeling real life situations

Learners answer the following questions

1. A school in Ndola is planning a cultural day. Each group contributes a fixed amount of k150 plus an extra amount depending on supplies. There are several groups of learners. Using x for the extra amount and g for the number of groups, write and simplify the expression for the total contribution.
2. A farmer in Chipata harvests 300 bags of maize. Each bag sells for a fixed price of k285 plus an amount that depends on market conditions.
 - a. Using m for the amount depending on market conditions, write and expand the expression for the farmer's total income.
 - b. If the demand for maize increases, would m increase or decrease? Justify your answer
3. A trader in Kitwe travels to the market several times a month. Each trip costs a fixed fee plus a variable fuel cost. Write and simplify the expression for monthly transport expenses.

4. A tailor in Mulobezi makes shirts and trousers each week. Each item has a fixed price (trousers k120, shirt k80) plus a variable cost depending on fabric quality. Given that the tailor made 7 trousers and 12 shirts in a particular week, write, expand and simplify the expression for the income that week.



Content tips:

- In simplifying expressions combine like terms after expansion.
- Reinforce that simplification makes expressions easier to interpret and use.
- To identify the variable learners should:
 - o Identify the Changing Quantities
 - o Clearly distinguish Fixed vs. Variable quantities. The fixed quantity is the constant.
- For instance, in activity 2.1 question 4, given that a learner selects v and w to represent variable cost depending on fabric quality for trousers and shirts respectively, the expression will be
 - o Each trouser: $(120 + v)$, where v is the variable cost depending on fabric.
 - o Each shirt: $(80 + w)$, where w is the variable cost depending on fabric.
 - o Tailor makes 7 trousers and 12 shirts. The expression can be written as follows:
$$7(120 + v) + 12(80 + w)$$
 - o This can then be simplified as

Expand:

$$(7 \times 120) + (7 \times v) + (12 \times 80) + (12 \times w)$$

Simplify:

$$1800 + 7v + 12w$$

Therefore, the income expression will be

$$1800 + 7v + 12w$$

Learning activity 3: Factorising Linear and Quadratic Expressions (common factor, grouping terms, difference of two squares, quadratic expression)



Activity 3.1: Discovering Common Factor Using Algebra Tiles

Materials needed:

- Algebra tiles (or paper squares/rectangles)
 - 'x-tiles' to represent (x)
 - 'unit tiles' to represent (1)
- Marker or chalk

Procedure:

- Step 1: place learners in small groups.
- Step 2: give each group x tiles and unit tiles

- Step 3: provide each group a card with a linear expression written on it (e.g $3x + 6$, $4x + 12$ **Note:** the expression should have two terms and should have a common factor)
- Step 4: let each group build an expression with tiles.
- Step 5: learners should group the tiles
- Step 6: ask learners to arrange the tiles into equal groups.
- Step 7: in the group's learners should answer the following:
 - i. What was the largest number of equal groups you could form?
 - ii. What does each group contain?
 - iii. What does the number of groups represent?
- Step 8: let learners explore different types of linear expressions using tiles (e.g $2x + 4y$, $5xy + 15$)



Content tips:

- Factorising by common factor means grouping terms into equal groups and finding the factor that the terms have in common.
- In activity 3.1 the learners should make equal groups. For instance, if the group was given $3x + 6$ They should make groups that each contain one x tile and 2 unit tiles
- 3 groups represent the common factor.
- Learners visualise that:

$$3x + 6$$

$$3(x + 2)$$

- when the two terms have different variables, ensure that you provide tiles that are in the expression [note : xy is one term hence you will have one tile representing xy]



Activity 3.2: non - linear expressions

Materials needed:

- o Algebra tiles (or paper squares/rectangles)
 - ' x^2 - tiles' to represent (x^2)
 - ' x - tiles' to represent (x)
 - '*unit tiles*' to represent (1)
- o Marker or chalk

Procedure:

Step 1: place learners in small groups.

Step 2: give each group ' x^2 - tiles' , ' x - tiles' .

Step 3: provide each group a card with a quadratic expression (quadratic expression with no constant) written on it (e.g $3x^2 + 6x$, $4x^2 + 12x$). Other variables can be used and learners given corresponding variable tiles.

Note: the expression should have two terms and should have a common factor

Step 4: let each group build an expression with tiles.

Step 5: group the Tiles

Step 6: ask learners to arrange the tiles into equal groups.

Step 7: In the groups learners answer the following:

- What was the largest number of equal groups you could form?
- What does each group contain?
- What does the number of groups represent?

Step 8: How can this grouping be expressed in mathematical terms?

Step 9: ask learners,

- do both terms still have x ?
- what does this x represent?
- Can this expression be further simplified?



Content tips :

- When factorising an expression where the terms have same variable but the power is different, begin by finding the common numerical factor and look for the common variable(s) in all terms then take the smallest power of that variable and factor it outside the brackets.
- For instance: in activity 3.2 if the learners were given $4x^2 + 12x$ they would factorise it as :
 - first find the numerical factor, by creating four groups of one x^2 tile and three x tiles.

- then have;

$$4(x^2 + 3x)$$

- The factorisation would not be complete as there is still a common variable x . Post factorizing x will give:

$$4(x + 3)x$$

As all the terms are connected by multiplication, this expression can be rewritten as

$$4x(x + 3)$$



Activity 3.3 :Factoring negatives

Give the learners the following for extended practice

- a. $-6x - 12x$
- b. $-8y - 4$
- c. $-10a - 15b$



Content tips:

- Factor out negatives if all terms are negative.



Activity 3.4: Factorising modeled expressions

Instructions:

- In pairs or small groups (3-4) learners answer the following questions and present their work.
1. A family spends $6x^2$ kwacha on mealie meal and $9x$ kwacha on vegetables in a week.
 - a. Write an expression for the total weekly expenditure.
 - b. Factorise the expression.
 - c. Explain what the common factor represents in the context of the family's shopping.
 - d. If the family shops for 4 weeks, how much will they spend in total on the common factor?
 2. A commuter spends $12a^2b$ kwacha on bus fares and $18ab^2$ kwacha on taxis in a month.
 - a. Write an expression for the total monthly transport cost.
 - b. Factorise the expression.
 - c. Explain what the factors represent in the context of transport expenses.
 3. A farmer spends $20x^3y^2$ kwacha on fertilizer and $25x^2y^3$ kwacha on seeds.

- a. Write an expression for the
 - b. Explain what the common factor represents in the context of the farmer's purchases.
 - (d) If the farmer repeats this purchase 3 times in a season, how much will he spend on the common factor?
4. A builder spends $15a^2b$ kwacha on cement, $20ab^2$ kwacha on sand, and $25ab$ kwacha on gravel.
- a. Write an expression representing the total cost of the items.
 - b. Factorise the expression.
 - c. Explain what the common factor represents in the context of the builder's materials.
 - d. If the builder buys these items twice, how much will he spend on the common factor?



Content tips

- Factorisation helps us identify common factors. In real life, these would be shared quantities or costs. We can interpret what they represent in practical terms (like shared shopping items, repeated transport costs, or building materials).

Forexample with the Family's Shopping given using the expression: $(6x^2 + 9x)$ factorized as $(3x(2x + 3))$. The common *factor* $(3x)$ represents the basic repeated unit of expenditure (a shared cost pattern in both mealie meal and vegetables).



Activity 3.5: "Rearrange to Reveal the Factors"

Instructions:

- Place learners in small groups.
- Give each group 4 small paper cards each with one term written on it. For instance, ax, ay, bx, by
- Learners to make an expression from the cards such as :

$$ax + ay + bx + by$$

- Learners to select two terms that have a common factor.
- Find the Common Factor in each group
- In groups learners to discuss:
 - Why did the first arrangement work better?
 - What made the second factor appear?
 - What happens if the terms are grouped differently?



Content tips:

- Factorization by grouping means splitting the expression into two groups of terms, factoring each group separately then identifying a common binomial factor that can be factored out.

For instance in activity 3.5 the terms ax, ay, bx, by from the expression

$$\mathbf{ax + ay + bx + by}$$

Can be grouped as:

$(ax + ay)$ and $(bx + by)$ alternatively

$$(ax + bx) \text{ and } (ay + by)$$

- Factorizing each group will yield :

Group 1:

$ax + ay \rightarrow$ factor **a**

Group 2:

$bx + by \rightarrow$ factor **b**

This will result into the expression

$$a(x + y) + b(x + y)$$

Which will have another common factor **(x + y)** so they factor again and obtain

$$(x + y)(a + b)$$



Activity 3.6: Factorisation Challenge

Instructions:

- Learners answer the following questions individually.
- 1. In each of the following, write an appropriate expression and factorise by grouping.
 - a. A family spent $(xy + 2x + 3y + 6)$ kwacha on mealie meal and vegetables.
 - b. A commuter spends $(ab + 2a)$ kwacha on bus fares and $(3b + 6)$ kwacha on taxis.
 - c. A farmer spends $(pq + 4p)$ kwacha on fertilizer and $(3q + 12)$ kwacha on seeds.
 - d. A builder spends $(mn + 5m)$ kwacha on cement and $(2n + 10)$ kwacha on sand.
 - e. A builder spends $(mn + 9m)$ kwacha on bricks and $(4n + 36)$ kwacha on timber.



Activity 3.7: Discovering the Difference of Two Squares

Instructions

- Divide learners into small groups and give each group the following worksheet.

Task 1: Investigate the Products and write the solutions in the table

Expression	Expanded form
$(a + 1)(a - 1)$	
$(x + 2)(x - 2)$	
$(y + 3)(y - 3)$	
$(x + 5)(x - 5)$	
$(2y + 5)(2y - 5)$	
$(xy + b)(xy - b)$	

Task 2:

- What do you notice about the first terms of the results?
- What happens to the middle terms when expanding?
- Why do the middle terms disappear?
- What do you notice about the numbers being subtracted?

Task 3: Predicting Rule

Ask learners to **predict the result** without expanding.

Task 4 : Reverse the Process

Now ask learners to work backwards.

Factorise:

- $x^2 - 16$
- $x^2 - 36$
- $x^2 - 64$

They use the pattern discovered.

Task 5 : Learner Conclusion

Each group writes the rule they discovered.

The expected conclusion is:

$$a^2 - b^2 = (a - b)(a + b)$$



Content tips:

- Difference of two squares factorization always factors into the product of a sum and a difference. It works only when both squares are perfect squares.
- An expression with two perfect squares $(a^2 - b^2)$ can be factorized as $(a - b)(a + b)$.

- From activity 3.5 Learners should notice that the result is always a square minus another square. Given an expression $(x + 7)(x - 7)$ learners will predict $x^2 - 49$.



Activity 3.8: Difference of two squares explored

Instructions

- In small groups learners factorize each of the following
 - a. $x^2 - 25$
 - b. $x^2 - 1$
 - c. $4a^2 - b^2$
 - d. $4x^2 - 16^2$
 - e. $2x^2 - 50$
 - f. $3 - 27x^2$
 - g. $-9m^2 + 49$



Content tips:

- If the difference of two squares isn't clear, factor out the highest common factor for all terms in the expression. For instance:

$$2x^2 - 50$$

$$2(x^2 - 25)$$

$$2(x^2 - (5)^2)$$

$$2(x - 5)(x + 5)$$

- If the expression is written in a way that does not show the difference of the terms, learners should rearrange the terms. For instance :

$$-9m^2 + 49$$

$$49 - 9m^2$$

$$7^2 - (3m)^2$$

$$(7 - 3m)(7 + 3m)$$



Activity 3.9: Difference of two squares illustration

Materials needed:

- Square paper or cardboard
- Ruler
- Scissors
- Pencil

Instructions

- Place the learners in small groups
- Learners to follow the given procedure

Procedure

Step 1: Construct Two Squares

- Cut two squares from paper or cardboard: A large square with side length a , a smaller square with side length b , where $a > b$
- Write expressions for the area of each square:

Step 2: Remove the Smaller Square

- Place the small square in one corner of the large square so that its sides line up with the sides of the large square. Trace its outline and cut it out.
- Write an expression for the area of the remaining shape

Step 3: Rearrange the Shape

- Cut off the rectangular part that remains along the side where the small square was removed.
- Move this cut piece and attach it to the opposite side of the remaining shape and form one rectangle.

Step 4: Describe the Rectangle

- Write an expression for the area of the rectangle.

Step 5: Compare the Areas

- What was the expression for the area after removing the small square?

- What is the expression for the area of the rectangle?

Step 6: Group Presentations

Each group presents:

- Their diagram of the squares
- The rearranged rectangle
- The expressions they obtained
- The relationship they discovered



Content tips:

- The area of the big square is a^2 and that of the small square is b^2
- Cutting off the small square from the large one leaves $a^2 - b^2$
- The area of the rectangle created is $(a + b)(a - b)$
- The area $a^2 - b^2$ has been expressed as $(a + b)(a - b)$, confirming the difference of two squares.



Activity 3.10: Discovering Factorisation of Quadratic Expressions ($ax^2 + bx + c; a \neq 0$)

Instructions:

- In small groups, learners answer the following questions

Step 1: Expand First

- learners to expand the following expressions:

Expression	Expanded Form
$(x + 2)(x + 3)$	
$(x + 4)(x + 5)$	
$(x + 1)(x + 6)$	
$(x + 3)(x + 7)$	
$(x + 5)(x - 2)$	
$(x + 6)(x - 1)$	
$(x - 2)(x - 3)$	
$(x - 4)(x - 5)$	
$(3x + 2)(x - 5)$	

Step 2: Pattern Investigation

- In groups, learners observe and discuss:
 - What do you notice about the coefficient of x^2 in every expansion?

- ii. How is the middle term related to the numbers in the brackets?
- iii. How is the constant and coefficient of x^2 related to the numbers in the brackets.

Step 3: Reverse the Process

Learners to factorise each of the following and state the steps used

- 1. $x^2 + 5x + 6$
- 2. $x^2 + 9x + 20$
- 3. $x^2 + 10x + 21$

Learners present their findings and justify their solutions



Content tips:

- The expansion of a binomial by binomial expression in the same variable (highest power 1) will produce a quadratic expression expressed in the form $ax^2 + bx + c$ where $a \neq 0$. For instance, in the activity 3.6, the expansion will be:

Expression	Expanded Form
$(x + 2)(x + 3)$	$(x^2 + 5x + 6)$
$(x + 4)(x + 5)$	$(x^2 + 9x + 20)$

Expression	Expanded Form
$(x + 1)(x + 6)$	$(x^2 + 7x + 6)$
$(x + 3)(x + 7)$	$(x^2 + 10x + 21)$
$(x + 5)(x - 2)$	$x^2 + 3x - 10$
$(x + 6)(x - 1)$	$x^2 + 5x - 6$
$(x - 2)(x - 3)$	$(x^2 - 5x + 6)$
$(x - 4)(x - 5)$	$(x^2 - 9x + 20)$

when $a = 1$ the quadratic expression can be factorized by finding two numbers that add to b (coefficient of x) and whose product is equal to ac (a - coefficient of x^2 and c is the constant. If these numbers are r and p the factorized form will be $(x + r)(x + p)$



Activity 3.11: Quadratic expressions factorization

Instructions:

- Place learners in small groups and have them answer the following questions

Task :

i. Complete the table below

Quadratic Expression	Product of a and c	Middle term	Factors
$2x^2 + 7x + 3$			
$2x^2 + 5x + 2$			
$3x^2 + 11x + 6$			

ii. In each of the quadratic expressions rewrite the middle term with the two factors found and factorize by grouping.

iii. A learner splits the middle term of $3x^2 + 11x + 6$ as:

$$3x^2 + 7x + 4x + 6$$

but is unable to factorise by grouping. Explain what mistake the learner made and why the grouping does not work.



Content tips:

- When the value of $a \neq 1$, the factors found can be used to split the middle term in order to factorise by grouping.
- The splitting of the middle term can be used even when $a = 1$.



Activity 3.12: Factorization

- Learners answer the following questions individually
- 1. Mr. Mubita's income after selling a certain number of pairs of uniform at a certain price. The total income was given using the expression $2x^2 + 7x + 3$. Factorise to find possible combinations of the price and number of uniforms sold.
- 2. A factory's production cost is given by $5p^2 - 13p + 8$. Factorise to find possible factors affecting cost.
- 3. A concert sells tickets in a way that the total revenue can be modelled by $4m^2 + 4m - 15$. Factorise to find the possible factors influencing ticket sales.

Learning Activity 4: Simplifying Algebraic Fractions



Activity 4.1: Exploring algebraic fractions

Instructions:

- In small groups learners answer the following questions.

1. A school club buys an unknown number of samosas to share equally among 5 learners. Write an expression of the number of samosas each learner gets?
2. A group of (n) learners hires a taxi that costs K120 for the trip. How much does each learner pay?
3. A cyclist travels (d) kilometres in (t) hours. What is the cyclist's average speed?
4. A class raises (k) kwacha selling snacks. The money is shared equally among (m) groups working on different projects. How much does each group receive?
5. A class prepared 120 buns for a school event. Initially, x learners were expected to attend. However, 3 more learners joined the group.
 - i. How many learners are now sharing the buns?
 - ii. How many buns does each learner receive?
6. A taxi charges K90 for a trip. Originally, x passengers planned to share the cost, but 2 more passengers joined before the taxi left.
 - i. How many passengers are sharing the fare now?
 - ii. How much does each passenger pay?
7. For each of the solutions to questions 1 to 6:
 - i. What do you notice about all the expressions you wrote?
 - ii. What does the top number represent?
 - iii. What does the bottom number represent?



Content tips:

- Algebraic fractions often represent sharing, rates, or averages.
- For instance in activity 4.1

- Question 1 learners choose a variable to express the number of samosas bought. If the variable is n the expression becomes $\frac{n}{5}$
- Question 5 the expression to be formed is $\frac{120}{x+3}$.



Activity 4.2: Expressing fractions in the lowest form

Instructions:

- In small groups learners answer the following questions

1. Simplify each of the following fractions

i. $\frac{20}{100}$

ii. $\frac{12}{20}$

iii. $\frac{3}{15}$

iv. $\frac{9}{30}$

2. What steps did you take to find your answer.
- Learners to present their solutions to the class



Content tips:

- Find the Highest Common Factor (HCF) of the numerator and the denominator. The HCF is the largest number that divides both the numerator and denominator exactly.
- Divide both numerator and denominator by HCF. This reduces the fraction to its simplest form.
- Write the simplified fraction.
- The steps used for numerical fractions are the same as those for algebraic fractions



Activity 4.3: Expressing algebraic fractions in the lowest form

Instructions:

- in small groups learners answer the following questions

1. A coach schedules $(18z^4)$ minutes of training spread across $(24z^6)$ sessions. How many minutes of training did each session last?
2. A teacher has $(30a^5)$ books and shares them equally among $(45a^7)$ students.
How many books did each student receive?
3. A farmer harvests $(14m^4)$ bags of maize and distributes them among $(21m^6)$ households. How many bags of maize did each household get?
4. A bus covers $(8p^3)$ kilometers using $(12p^5)$ liters of fuel. How many kilometers were covered per liter of fuel?
5. A factory produces $(27q^6)$ items in total, packed into $(36q^9)$ boxes. How many items were packed into each box?
6. A gardener uses $(x^2 + 5x + 6)$ liters of water for plants, spread across $(x^2 - 9)$ flower beds. How much water did each flower bed receive?

**Content tips:**

- Algebraic fractions are simplified in the same way as numerical fractions.
- For instance activity 4.3
- Question 1

$\frac{18z^4}{24z^2}$ Factorise the numerator and denominator by the HCF $6z^2$

$\frac{6z^2 \times 3z^2}{6z^2 \times 4}$ This will give us

$$\frac{3z^2}{4}$$

- When the factor is not clear, factorise the numerator and denominator. For instance in Question 6

$\frac{x^2+5x+6}{x^2-9}$ Factorise the numerator and denominator

$\frac{(x+2)(x+3)}{(x+3)(x-3)}$ This will give us an HCF of $x + 3$

$$\frac{x+2}{x-3}$$

**Activity 4.4:** Fractions simplified (addition and subtraction)

Instructions:

- In small groups learners answer the following questions

1. Add/ subtract the following fractions

i. $\frac{2}{5} + \frac{1}{5}$

ii. $\frac{3}{7} - \frac{2}{7}$

iii. $\frac{2}{3} + \frac{1}{12}$

iv. $\frac{2}{3} + \frac{1}{5}$

v. $\frac{7}{6} - \frac{3}{7}$

2. What steps did you take to find your answer.

- Learners to present their solutions to the class



Content tips:

- When the denominators of the fractions are the same : Add or subtract the numerators and keep the denominator the same
- When the denominators are different: proceed as follows

$$\frac{2}{3} + \frac{1}{5} \quad \text{the Lowest Common Denominator (LCD) is 15}$$

the first fraction will become $\frac{10}{15}$ while the second fraction will become $\frac{3}{15}$

the numerators can now be added to get $\frac{13}{15}$

- Algebraic fractions follow the same rules as numerical fractions



Activity 4.3 :

Instructions

- In small groups learners answer the following questions
 - Given that algebraic fractions follow the same rules as numerical fractions answer the following questions:
 - a. Scenario: At a school, learners are sharing snacks during break time. Let x represent the number of learners sharing a snack. Given that a loaf of bread has 3 pieces and a chocolate bar has 5 pieces:
 - i. How many pieces of bread will each learner receive?
 - ii. How many pieces of chocolate will each learner receive?
 - iii. If a gift bag was made for each learner how many pieces of both snacks altogether will each gift bag have?
 - b. **Scenario:** At a school fundraising event, two classes sell bottled drinks. The profit each class makes depends on the number of bottles sold.
 - Class A makes a profit represented by

$$\frac{3x + 5}{x - 1}$$

Class B makes a profit represented by

$$\frac{2x + 4}{x - 1}$$

-

where (x) represents the number of boxes of drinks sold.

- i. Write an algebraic expression that represents the total profit made by both classes.
 - ii. Simplify the expression into a single algebraic fraction.
 - iii. State the value of (x) that would make the expression undefined and explain why.
- c. A farmer uses $\frac{2}{x}$ of land for maize $\frac{1}{x+1}$ of land for beans. What fraction of land is used altogether?
- d. Scenario: During a school event juice is shared among learners. In the first class, x liters of juice are shared equally among 4 learners. In the second-class x liters of juice are shared equally among 6 learners.
- i. Write down the fraction representing the amount of juice each learner gets in each class.
 - ii. If two learners, one from each class combines their juice, how much juice do they have altogether.
- e. Scenario: Two classes contribute money for a school trip. Class A contributes r kwacha shared among r + 6 learners. Class B contributes r Kwacha shared among r + 3 learners. How much more does one learner in class A receive compared to class B.

- f. A baker uses $\frac{2}{x+1}$ cups of sugar for one cake, while another baker uses $\frac{3}{x+2}$ cups for a different cake. What is the total number of cups of sugar used by the two bakers?



Content tips:

- Identify the operation: Check if the problem requires addition or subtraction.
- Factor the denominators (if possible): Check whether the denominators can be factored. This helps when finding the common denominator.
- Find the Lowest Common Denominator (LCD): Identify the smallest expression that both denominators can divide into.
- Rewrite each fraction with the LCD: Multiply the numerator and denominator of each fraction by the necessary expression so that both fractions have the same denominator.
- Add the numerators: Once the denominators are the same, add the numerators while keeping the denominator unchanged.
- Simplify the result



Activity 4.4: Fractions simplified (multiplication and division)

Instructions:

- In small groups learners answer the following questions

Work out each of the following fractions

i. $\frac{4}{7} \times \frac{3}{10}$

ii. $\frac{3}{7} \times \frac{2}{7}$

iii. $\frac{5}{6} \times \frac{2}{7}$

iv. $\frac{2}{3} \div \frac{5}{7}$

v. $\frac{1}{3} \div \frac{5}{12}$

iv. Explain the steps you used to find the answer.

- Learners to present their solutions to the class to consolidate their findings.



Content tips:

- When multiplying fractions:
 - o Multiply the numerators together.
 - o Multiply the denominators together.
 - o Simplify the fraction if possible.
- When dividing fractions:
 - o Change the division sign to multiplication.
 - o Find the reciprocal (flip) of the second fraction.
 - o Multiply the fractions.
 - o Simplify the result if possible.

- We change the division into multiplication because multiplication and division are inverse operations. To maintain the correct result, we take the reciprocal (flip) of the divisor. Multiplying by the reciprocal is easier than dividing fractions directly.
- Algebraic fractions follow the same rules as numerical fractions when multiplying or dividing.



Activity 4.5: Fractions simplified (multiplication and division)

Instructions:

- In small groups learners answer the following questions:

1. Each solar panel produces $\frac{x^2}{y^3}$ units of energy. How many units of energy will be generated by $\frac{y^2}{z^3}$ solar panels?
2. A household has a router with $\frac{2m^3n^2}{p^3}$ GB of internet data. If it can only produce $\frac{3m^2n^2}{4p^2}$ GB per device. How many devices can be supported with the available data? Simplify your answer.
3. A factory produces $\frac{6a^2}{b^3}$ items per hour. How many items will be produced in $\frac{b^2}{c^2}$ hours?
4. A recipe requires $\frac{5x}{y^2}$ liters of milk per serving. How many servings can be made with $\frac{20x}{y^3}$ liters of milk?



Content tips:

- Identify the operation: check if the problem requires multiplication or division.
- The steps used for numerical fractions are the same for algebraic fractions.
- For instance in activity 4.5:

$$\frac{x^2}{y^3} \times \frac{y^2}{z^3}$$

$$\frac{x^2 \times y^2}{y^3 \times z^3} \quad \text{simplify the fraction}$$

$$\frac{x^2}{yz^3}$$

- For division from the same activity 4.5 question number 2 .

$$\frac{2m^3n^2}{p^3} \div \frac{3m^2n^2}{4p^2}$$

$$\frac{2m^3n^2}{p^3} \times \frac{4p^2}{3m^2n^2}$$

$$\frac{2m^3n^2 \times 4p^2}{p^3 \times 3m^2n^2}$$

$$\frac{8m^3n^2p^2}{3p^3m^2n^2}$$

$$\frac{8m}{3p}$$

Expected Standard: Algebraic expressions applied in problem solving correctly

TOPIC 4: POLYGONS

Introduction:

Polygons surround us in ways both obvious and subtle. They are two-dimensional shapes that form surfaces of three-dimensional shapes such as pavers, road signs, windows and in the intricate geometric patterns woven into traditional fabrics and crafts passed down through generations. These shapes formed by straight lines meeting at angles are more than mere drawings on paper; they represent order, symmetry, and structure in the world around us



Pavers



Road Signage



key terms: Regular polygon, Irregular polygon, Interior angle, Exterior angle

General competences: Analytical Thinking, Critical Thinking, Problem Solving, Creativity, Communication, Collaboration, Digital Literacy



key question



Where do we see polygons
shaping the world around us?

Sub-Topic 1: Regular and Irregular Polygons

Introduction:

This subtopic explores **regular polygons** and **irregular polygons**, focusing on their **main properties and how they appear in everyday life**.



regular polygons
on soccer ball
surface



irregular polygon on a
surface of a building

Specific competence: Create different objects using polygons

Learning activity 1: *Exploring Different Types of Polygons (Regular and Irregular)*



Activity 1.1: Classifying polygons

Materials Needed

- Cut outs of polygons (both regular and irregular)
- Rulers and protractors

- Chart paper or poster boards
- Markers, pencils, and erasers

Optional: tablets or computers for digital exploration

Instructions:

- Place learners in small groups
- Provide each group with the materials needed.

Procedure: each group to:

- make observations on the shapes given
- use their observations to group the polygons
- make a presentation explaining why they grouped the polygons the way they did.



Content Tips:

- Learners to group the polygons into regular and irregular
- Regular Polygons have:
 - **Equal sides:** All sides are the same length.
 - **Equal angles:** Each interior angle is the same size.
 - **Symmetry:** Regular polygons have multiple lines of symmetry
- Irregular Polygons have

- **Unequal sides:** At least two sides are of different lengths.
- **Unequal angles:** Interior angles are not all the same.
- **Symmetry:** Most irregular polygons lack symmetry



Activity 1.2 Polygons in Everyday Designs

Materials Needed:

Graph paper or plain paper

Rulers and protractors

Coloured pencils/markers

Optional: tablets/computers for digital drawing

Instructions:

- Divide learners into small groups.
- Each group selects a real-life object or setting where polygons are used
- Groups draw their chosen object on paper and must:
 - Include at least two regular polygons (e.g., square, equilateral triangle, regular hexagon).
 - Include at least two irregular polygons.
 - Clearly label each polygon in their design.
- Groups prepare an explanation describing:
 - i. Which shapes are regular polygons



- ii. Which shapes are irregular polygons
- iii. Why those shapes might be useful in the real-life objects

- Each group presents their design to the class, highlighting the role of polygons in real-world applications.



Content tips

- The task can be assessed using the rubric illustrated

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Identification of Polygons	Clearly identifies and correctly uses both regular and irregular polygons in the design.	Identifies and uses both types of polygons with minor errors.	Uses polygons but shows limited understanding of regular and irregular polygons.	Shows little or no understanding of the types of polygons used.
Design of Real-Life Object	Design is clear, creative, and strongly represents a real-life object or setting.	Design represents a real-life object with some creativity.	Design shows some connection to a real-life object but lacks clarity.	Design is unclear or does not represent a real-life object well.
Application of Polygons	Demonstrates clear understanding of how polygons are used in real-life structures or	Shows some understanding of how polygons	Limited explanation of how polygons relate to the object.	No clear connection between polygons and the real-life object.

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
	objects.	apply in real life.		
Group Collaboration	All group members participate actively and work well together.	Most group members participate.	Some members participate while others contribute little.	Very little cooperation within the group.
Presentation and Explanation	Presentation is clear, confident, and explains the role of polygons effectively.	Presentation is understandable with some explanation of polygons.	Presentation is brief or lacks clear explanation.	Presentation is unclear or incomplete.

Scoring Guide

- **16–20 points:** Excellent understanding
- **11–15 points:** Good understanding
- **6–10 points:** Satisfactory understanding
- **1–5 points:** Needs improvement

Learning activity 2: Exploring the sides and angles of a polygon



Activity 2.1. Stick and Joint Activity (Building Polygons)

Materials:

1. Sticks or straws (straight, same size)
2. Clay, plasticine

Instructions:

- Learners build shapes using sticks joined at the ends (such as triangle, quadrilateral, pentagon ...)
- Each joint represents a corner (angle).
- Learners answer the following questions
 - i. How many sticks did you use?
 - ii. How many corners did the shape form?
 - iii. What do you notice about the number of sides and corners?

Polygon name	Number of Sides (n)
1.	
2.	
3.	
4.	



Activity 2.2.: Angle and side Exploration

Instructions

- Place learners in small groups
- Learners draw polygons of their choice and state if it is regular or irregular
- Extend one side on each vertex to form an exterior angle. Learners to measure the interior and exterior angle at any point.
- They then fill in the table below

Polygon	Number of Sides (n)	Interior Angle	Exterior angle	Sum of interior angles	Sum of exterior angles

- From the data in the table what observations can you make about:
 - i. the interior and exterior angles sum?
 - ii. sum of the exterior angles?

- iii. size of interior angles and exterior angles as the number of sides increases?



Content tips:

From the two activities learners observe:

- In any polygon, the number of sides is equal to the number of angles.
- At each vertex, the interior angle and its adjacent exterior angle add up to 180° .
- The sum of exterior angles of any polygon is always 360° .
- As the number of sides increases, the size of the interior angles increase, while the size of the exterior angles decrease.

Learning activity 3 Calculating the Interior and Exterior Angles of a Polygon



Activity 3.1: Pattern discovery

Instructions:

- From Activity 2.2 in small groups learners to observe the connection between exterior angles and interior angles. Using observations, learners explain how they can calculate the interior angle given the exterior angles or vice versa.



Content tip:

- Learners will observe that:

$$\text{interior angle} + \text{exterior angle} = 180^\circ$$

$$\text{interior angle} = 180^\circ - \text{exterior angle}$$

$$\text{exterior angle} = 180^\circ - \text{interior angle}$$



Activity 3.2: Interior and Exterior angles in real life

Instructions:

- Learners to analyse and answer questions individually.
1. A community art group in Lusaka is designing a decorative tiled platform shaped like a regular polygon for a public park. While laying the tiles around the edge, the workers notice that each turning angle they make at every corner is 30° .
 - a. What is the size of each interior angle of the platform?
 - b. How many sides does the platform have?
 - c. What is the total sum of all the interior angles?
 - d. Without calculating, decide which platform would have more sides:

- i. One with a turning angle of 45°
- ii. One with a turning angle of 20°
- iii. Explain your thinking.

- 2. A floor tiler is laying tiles with a surface in the form of an irregular heptagon in a classroom. Each tile has one interior angle of 120° . What is the measure of the exterior angle at that corner of a tile?
- 3. A traffic engineer is designing a stop sign, with a surface in the shape which is a regular octagon. Each interior angle is 135° . Find the measure of each exterior angle of the stop sign.
- 4. A local art center is creating a patterned frame with a surface in the shape of a regular polygon. Each exterior angle of the frame is 45° . What is the measure of each interior angle of the frame?
- 5. A decorative window grill is designed with a surface in the shape of a regular polygon with interior angles of 150° . What is the measure of each exterior angle of the grill?

Learning activity 4: Determining the Sum of Interior angles of a Polygon



Activity 4.1: Finding the sum of interior angles of polygons

Materials Needed

Plain paper or chart paper

Rulers and pencils

Protractors (for measuring angles)

Colored markers

Instructions:

- Place learners in small groups (3–5 learners).
- Learners to do the following:
 - draw polygons with different number of sides (3 sides to 7 sides)
 - draw diagonals from one vertex, to divide the polygon into triangles.
 - record the following information from their drawings in a table like the one shown below

polygon	Number of sides	Number of triangles	Sum of interior angles

- Ask the learner what they have observed:
 - in terms of the number of sides and the number of triangles they drew in a particular polygon.
 - the sum of interior angles of a particular polygon (by measuring) and the total sum of angles of the triangles in the polygon



Content tips:

- Learners should observe the following:
- The number of triangles in the polygon is 2 less than the number of sides
- The sum of interior angles of a polygon is equal to the sum of angles of the triangles in the polygon
- Sum of interior angles is;

$$(n - 2)180^\circ$$

Where $(n - 2)$ is the number of triangles and 180° as a sum of interior angles of each triangle



Activity 4.3: Interior angles of regular polygons

Instructions

- Learners complete the table below with any four different regular polygons of their choice and answer the following questions

polygon	Number of sides	Size of interior angle	Sum of interior angles

- i. Observe the number of sides, size of interior angle and sum of interior angles:
 - a. Determine how we can get the size of one interior angle from the sum of interior angles and the number of sides.
 - b. Determine how we can find the number of sides of a regular polygon given the sum.
 - c. Determine how we can find the size of an exterior angle of a regular polygon.



Content tips:

- For regular polygons:

$$\text{size of an interior angle is} = \frac{\text{sum of interior angles}}{n}$$

$$\text{Number of sides is } (n) = \frac{\text{sum of interior angles}}{180} + 2$$

$$\text{Exterior angle is} = \frac{360^\circ}{\text{number of sides}}$$



Activity 4.4 : Angles in a polygon

1. A landscape designer is creating a regular hexagon-shaped flower bed. Each corner of the flower bed forms an interior angle. A regular hexagon has 6 sides.
 - i. Find the sum of its interior angles.
 - ii. Calculate each interior angle.
 - iii. Calculate each exterior angle.
 - iv. If the designer decided to make the garden larger but kept the same hexagon shape, would the interior angles change? Explain.

2. A decorative floor tile is shaped like a regular polygon. The angle formed at each corner *outside* the tile (when extending the sides) is 30° .
 - i. How many sides does the polygon have?
 - ii. What is the measure of each interior angle?
 - iii. Why does knowing just one exterior angle allow you to determine the number of sides?

3. A company designs a logo shaped like a regular polygon where each inside corner angle measures 144° .
 - i. What is the measure of each exterior angle?
 - ii. How many sides does the polygon have?
 - iii. Explain why the interior and exterior angles of a regular polygon must always add up to 180° .

4. A craftsman builds a decorative 12-sided (regular) clock frame.
 - i. Find the sum of its interior angles.
 - ii. Calculate each interior angle.
 - iii. Calculate each exterior angle.

5. Engineers design a road junction made of equal straight road sections forming a regular polygon surface. The turning angle at each corner (exterior angle) is 24° .
- How many sides does the polygon have?
 - What is the measure of each interior angle?
6. A math club designs a badge shaped like a regular polygon. They discover that the interior angle is five times the exterior angle.
- Write an equation to represent this relationship.
 - Solve to find the size of the exterior angle.
 - Find the size of the interior angle.
 - Determine the number of sides of the polygon.

Learning activity 5: using properties of polygons, in creating objects.



Activity 5.1 project – actualizing the design

- In activity 1.2 learners created a design using polygons. In this activity, using the same groups, learners should
 - o Make the actual object or model of the design they made in the mini project

- Present their creations to the class, explaining how polygon properties were used.



Content tips:

- Ensure that learners are aware of the rubric as they do the project.

Criteria	Level 4 — Excellent	Level 3 — Good	Level 2 — Developing	Level 1 — Beginning
Model Construction	Model is neat, strong, accurate, and closely matches the original design	Model is complete and mostly accurate with minor flaws	Model is incomplete or loosely matches design	Model is poorly constructed or not finished
Use of Polygons	Multiple polygons used correctly; at least one regular polygon clearly shown	More than one polygon used correctly	Limited variety of polygons used	Polygons used incorrectly or unclear
Mathematical Accuracy	Polygon names, sides, angles, and properties are all correct	Minor mathematical errors	Several errors in properties or measurements	Major misunderstandings of polygon properties
Application of Properties	Clear and correct explanation of how angle sums, symmetry, and sides	Good explanation with minor gaps	Basic explanation with limited links to properties	Little or no connection to polygon properties

Criteria	Level 4 — Excellent	Level 3 — Good	Level 2 — Developing	Level 1 — Beginning
	influenced design			
Creativity & Functionality	Design is original, practical, and visually appealing	Design is clear and somewhat creative	Design shows minimal creativity	Design lacks effort or purpose
Teamwork & Participation	All members actively contributed and collaborated well	Most members participated	Uneven participation	Little teamwork evident
Presentation Skills	Clear, confident, well-organized explanation; correct math language used	Mostly clear explanation; minor language errors	Explanation unclear or poorly organized	Unable to explain design clearly



Activity 5.2 : Project- 'Inter concept connection'

Instructions:

- In groups learners perform the following project
- Learners should be given enough time to complete the project
- Do not limit the task to the materials listed, use locally available materials and adapt the project to school/ community specifications.

Projects outline

"You are a team of landscape designers. Your task is to **design and build a small school garden** using different polygon-shaped sections for planting."

Materials: String or rope, Pegs/sticks, Measuring tape, Protractor, Chalk or lime powder (for marking ground), Graph paper (for planning).

Project Procedure

Part 1: Planning the Garden (On Paper)

- Draw a coordinate grid with appropriate scale, using the first quadrant only (e.g., 1 unit = 1 meter).
- Design your garden layout including:
 - At least **4 garden beds (polygons)**:
 - One **regular polygon** (e.g., square, hexagon)

- One **triangle**
 - One **irregular (non-regular) polygon**
 - One polygon with **more than 4 sides**
- iii. Label all vertices with coordinates.
 - iv. Calculate **perimeter** to estimate fencing needed
 - v. Calculate **area** to estimate soil required

Part 2: Transferring to the Ground

- i. Choose an open space at school.
- ii. Mark an origin point (0,0).
- iii. Use measuring tape to locate and mark coordinates using pegs.
- iv. Join the pegs using rope to form each polygon.
- v. Calculate actual **perimeter** built
- vi. Calculate actual **area**

Part 3: Angle Investigation

For each polygon:

- i. Calculate the **expected angle sum**

- ii. Measure actual angles using a protractor (or estimate using constructed corners)
- iii. Record all results in a table

Part 4: Error Calculation

For each polygon, calculate:

- i. Absolute error of the perimeter
- ii. Percentage error of the perimeter
- iii. Percentage error of the angle sum

Part 5: Garden Reflection and Improvement

Learners discuss and write:

- i. Which shape was easiest to construct on the ground? Why?
- ii. Which polygon had the largest error?
- iii. What caused the errors? (e.g., rope movement, uneven ground, inaccurate measuring)
- iv. How can the garden layout be improved?
- v. Why is accuracy important in real-life construction?
- vi. If you have the same amount of land, which shape of garden bed would give you the most efficient use of space and why?



Content tips:

- Adapt the project to suit locally available materials.
- Learners will apply knowledge from several concepts.
- The following rubric can be used to assess learners' competence levels.

Assessment Rubric

Criteria	Excellent (4)	Good (3)	Basic (2)	Emerging (1)
Planning (coordinates)	All points plotted correctly on the coordinate grid; coordinates labeled clearly; layout matches design	Most points plotted correctly; minor labeling mistakes or slight misplacement	Several points missing or incorrectly plotted; incomplete labeling; layout partially matches design	Very few or no points plotted; labeling missing; layout not followed
Construction	All polygons constructed accurately; sides measured correctly; shapes match planned coordinates	Most shapes constructed correctly; minor errors in side lengths or shape alignment	Shapes partially constructed; multiple side lengths or angles incorrect; not all shapes complete	Shapes not built or completely inaccurate
Angle work	All angles measured/calculated correctly; all polygons	All polygons analyzed but 1–2 small mistakes in angle	Some angles measured/calculated; some polygons missing; some calculations incorrect	Very few or no angles measured; calculations

Criteria	Excellent (4)	Good (3)	Basic (2)	Emerging (1)
	analyzed; sum of angles verified	measurements or sums		missing or completely wrong
Error calculation	All percentage errors calculated correctly; clear explanation of causes of error	Most errors calculated correctly; minor calculation mistakes or explanation incomplete	Attempted error calculation but incomplete or partially incorrect; explanation missing or unclear	Errors not calculated or completely wrong; no explanation
Collaboration	All members are actively involved throughout the task helps resolve problems respectfully	Most members participate; one may be less involved	Only a few members do most of the work	One or two learners dominate; others are not involved

Expected standard: different objects created using properties of polygons used to create different objects appropriately

References

Ministry of Education (2023) **Zambia Education Curriculum Framework**. Zambia: Lusaka, Curriculum Development Centre.

Ministry of Education (2024) **Mathematics I Syllabus- Ordinary Level Secondary Education Form 1-4**. Zambia: Lusaka, Curriculum Development Centre.

BOYD C. J etal (1999) **Mathematics' Application and connections course 1 and 2**, New York : Glencoe/Mcgraw- Hill.